Energy 347 (2026) 140315

Contents lists available at ScienceDirect

Energy

journal homepage: www.elsevier.com/locate/energy

Integrated CFD modeling and experimental validation of single-phase immersion cooling for data center thermal management

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ARTICLE INFO

Keywords:
Single-phase immersion liquid cooling data center
Heat transfer performance
Heat sink optimization
Computational fluid dynamics
Heat dissipation coefficient

ABSTRACT

Driven by the escalating power densities of modern servers, single-phase immersion cooling has emerged as a critical thermal management paradigm. However, the absence of a systematic methodology for selecting optimal coolants and components remains a significant challenge. This study addresses this gap by proposing a holistic framework that integrates theoretical analysis, computational fluid dynamics (CFD) simulations, and experimental validation. First, the flow and heat transfer characteristics of oil-based and fluorocarbon-based coolants were scrutinized through boundary layer theory and the field synergy principle. Concurrently, a novel Heat Dissipation Coefficient (HDC) was formulated to quantitatively evaluate coolant performance. Second, the thermo-hydraulic behavior of a standard 1U server was rigorously evaluated via CFD simulations and validated on a dedicated experimental platform. Building upon this, heat sink geometries were optimized using a hybrid CFD-Genetic Algorithm (GA) approach. Third, a novel tank architecture was engineered to mitigate flow maldistribution. The Realizable k-ε model and the SST k-ω model were respectively employed for server flow simulations and ribbed structure optimization. The results demonstrated that the HDC index provided a high-fidelity assessment of coolant performance, aligning closely with CFD data while significantly reducing computational overhead. Comparative analysis revealed that fluorocarbon coolants exhibited superior thermal performance over silicone oil. The optimized heat sink achieved a Performance Evaluation Criterion (PEC) of 2.05, while the redesigned tank structure reduced flow maldistribution by 85%. Finally, a 1000-h compatibility test confirmed the long-term operational reliability of the SS110 coolant.

1. Introduction

Data centers (DCs) represent sophisticated technological ecosystems comprising distributed server arrays, advanced networking, and critical infrastructure. They serve as the indispensable backbone for global data storage, transmission, and processing. Propelled by breakthroughs in large-scale integrated circuits and micro-nano manufacturing, transistor density has scaled dramatically, precipitating a surge in heat flux at the chip level. Currently, the Thermal Design Power (TDP) of mainstream processors has reached 280 W, while high-performance accelerators frequently exceed 700 W [1]. This escalating power density poses unprecedented thermal management challenges, necessitating advanced cooling paradigms to ensure operational reliability and energy sustainability. Globally, DC energy consumption is projected to surpass 1000 TWh by 2026 [2]. Conventional air-cooling systems, reliant on room-scale Computer Room Air Conditioning (CRAC), suffer from inherent inefficiencies due to significant airflow mixing losses and low heat capacity [3]. In such facilities, cooling infrastructure often accounts for 40% of the total operational expenditure, representing the largest energy overhead after IT equipment [4]. Consequently, thermal management has become the primary bottleneck in DC evolution, mandating the transition toward high-efficiency liquid cooling technologies.

Prior research has explored various strategies to mitigate DC energy consumption. Zhou et al. [5] demonstrated that elevating supply air temperatures could reduce energy demand. However, this often compromised hardware reliability. In contrast, liquid coolants offered significantly higher heat transfer coefficients due to their superior thermophysical properties [6]. Furthermore, liquid cooling facilitated

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https://doi.org/10.1016/j.energy.2026.140315
Received 24 October 2025; Received in revised form 26 January 2026; Accepted 2 February 2026
Available online 3 February 2026

high-grade waste heat recovery, enhancing overall energy circularity [7]. Jiao et al. [8] proposed a hybrid liquid-cooling architecture and optimized its structure via multi-objective algorithms. Nevertheless, spatial constraints within compact server chassis remained a significant implementation hurdle.

Current liquid-cooling solutions are categorized into four primary archetypes: indirect water cooling, direct-to-chip (cold plate) cooling, single-phase immersion cooling (SPIC), and two-phase immersion cooling (TPIC) [9]. Unlike water cooling systems, which are prone to leakage and fluid path blockages, immersion cooling provides a more robust and holistic thermal environment. By submerging the entire server in a dielectric fluid, immersion cooling enables direct heat extraction from all electronic components while shielding hardware from airborne contaminants, mechanical vibrations, and humidity [10]. The efficiency of these systems is typically benchmarked by Power Usage Effectiveness (PUE). Notably, micro-modular immersion systems, such as those pioneered by Green Revolution Cooling (GRC), can achieve PUE values as low as 1.03, vastly outperforming air-cooled and cold-plate-based counterparts [11]. Within the immersion domain, TPIC systems leverage the latent heat of vaporization to achieve exceptionally high heat transfer coefficients [12]. Ma et al. [13] further enhanced boiling heat transfer using sintered copper powder porous structures. However, applying this sintering process directly onto actual chip surfaces was challenging. Moreover, the adoption of TPIC is currently hindered by high capital expenditure and stringent requirements for hermetic sealing [14].

Compared to the aforementioned solutions, SPIC systems offer a compelling balance of high heat transfer efficiency, operational stability, and minimal acoustic noise [15]. The fundamental feasibility of SPIC has been validated using various fluids, including Novec 7100 [16], mineral oil [17], and Noah3000D [18]. Nevertheless, thermal performance is highly sensitive to coolant viscosity and flow dynamics. For instance, Gajjar et al. [19] identified the thermal superiority of FC-3283 over mineral oil, while Hnayno et al. [20] observed a 6% performance degradation when coolant viscosity increased from 4.6 to 9.8 mPa s.

Recent research has shifted toward structural and flow architecture optimization. Investigations into flow orientation revealed that counter-gravity configurations offer enhanced heat dissipation [21], while T-configuration layouts can achieve a 12.6% reduction in thermal resistance compared to Z-configurations [22]. Internal components such as baffle angles [23], aperture geometry [24], and the gap spacing between heat sinks and tanks [25] significantly influence flow uniformity and bypass effects. Specifically, minimizing server spacing [26] and maintaining low coolant flow rates [27] have been identified as effective strategies for ensuring safe operating temperatures. Furthermore, hybrid approaches, including the integration of micro-pumps [28,29] or combining cold plates with SPIC [30], have shown potential in reducing equivalent thermal resistance. From an energy perspective, pump-driven SPIC systems can reduce average coolant temperatures by 55.5% and improve PUE by 11.6% [31], starkly contrasting with traditional water-cooling systems whose PUE often exceeds 1.2 under high ambient temperatures [32]. Additionally, advanced modeling techniques, including gray-box dynamical models [33] and CFD-based parametric studies [34–36], have been established to capture the complex energy-mass transfer dynamics within these systems. While metrics like the Mouromtseff number have been used to rank coolants [37], their derivation from simple pipe-flow theory limits their accuracy in the complex geometries of high-density servers. Moreover, previous studies have often focused on average temperatures, neglecting the spatial maldistribution of the coolant and its impact on localized hotspots. Specifically, the CPU, characterized by its extreme power load, requires more rigorous monitoring as it is uniquely susceptible to thermal failure under suboptimal immersion.

While integrating heat pipes or cold plates into SPIC systems effectively reduces thermal resistance, such configurations inevitably increase structural complexity and potential leakage risks. Alternatively,

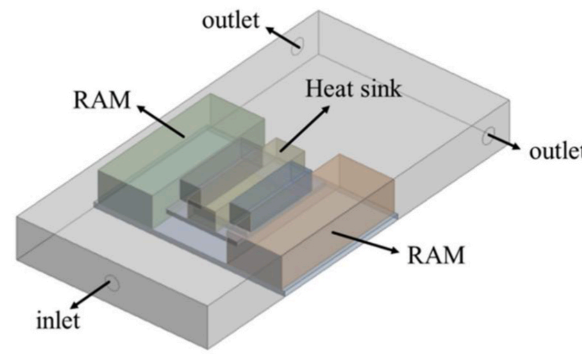

Fig. 1. Schematic diagram of immersion cooling server.

CFD-guided parametric optimization of heat sinks presents a pragmatic solution to enhance thermal performance without necessitating auxiliary hardware. Despite these advancements, critical research gaps persist in the context of SPIC systems. First, specialized evaluation metrics tailored for server thermal management scenarios are notably lacking. Current metrics, typically derived from internal pipe flow, have unverified applicability to the complex thermo-fluid dynamics within a server. Second, there remains significant potential to enhance the heat transfer capacity of existing coolants without escalating system complexity. Third, insufficient attention has been directed toward internal flow characteristics within immersion tanks, particularly the critical issue of coolant maldistribution. To bridge these gaps, this study presents four distinct contributions. First, we developed a comprehensive performance evaluation framework for SPIC coolants that rigorously accounted for their thermal-hydraulic characteristics. Second, we conducted systematic heat transfer experiments comparing SS110 coolant and silicone oil, demonstrating that SS110 achieved a temperature reduction of 16.5 °C and a 65% decrease in temperature non-uniformity. Third, we conducted a multi-objective optimization of the heat sink, culminating in a design with a PEC of 2.05. Fourth, we proposed a novel tank configuration featuring a symmetric inlet and outlet arrangement combined with a porous plate of non-uniform porosity. This approach significantly mitigated coolant maldistribution, achieving an 85% reduction.

2. Numerical methodology and experimental setup

2.1. Description of the model

A systematic evaluation of the thermo-hydraulic performance of diverse coolants was conducted within a 1U SPIC server framework, as illustrated in Fig. 1. The computational domain, comprising the processor, heat sink, random access memory (RAM) modules, and the PCB, was constructed in accordance with the architectural specifications established by Muneeshwaran et al. [21]. To ensure high predictive accuracy, three-dimensional conjugate heat transfer modeling was implemented and subsequently benchmarked against experimental datasets.

Numerical simulations were performed using ANSYS Fluent. To accurately resolve near-wall gradients and flow separation, the Realizable k-ε turbulence model was adopted in conjunction with an enhanced wall treatment scheme. The conservation equations for mass, momentum, and energy, alongside the turbulence transport equations, were defined as follows:

$$\nabla \cdot (\rho \boldsymbol{U}) = 0 \tag{1}$$

$$\nabla \cdot (\rho \boldsymbol{U}\boldsymbol{U}) = -\nabla p + \mu_{\text{eff}} \Delta \boldsymbol{U} \tag{2}$$

$$\nabla \cdot \left(\rho c_{\text{p}} \boldsymbol{U} T\right) = \nabla \cdot (\lambda \nabla T) + S_T \tag{3}$$

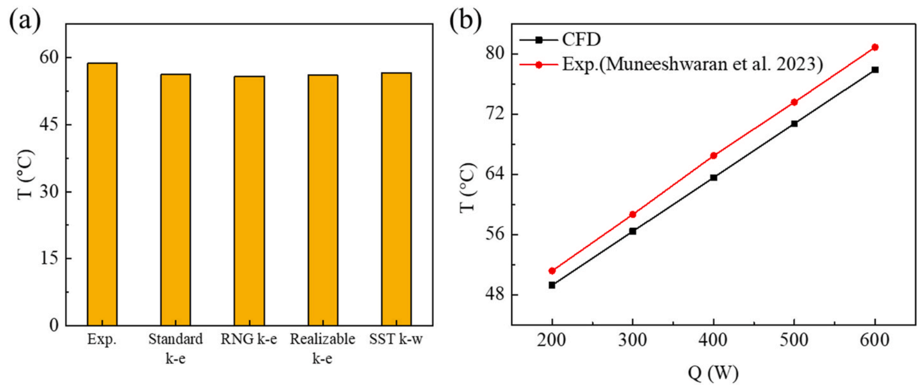

Fig. 2. Comparisons between numerical results and experimental data. (a) Effect of the turbulence model (b) Effect of heat generation rate.

$$\nabla \cdot (\rho \boldsymbol{U} k) = P_k - \rho\varepsilon + \nabla\cdot((\mu + \mu_t / \sigma_k)\nabla k) \tag{4}$$

$$\nabla \cdot (\rho \boldsymbol{U} \varepsilon) = C_1 \varepsilon S + \frac{\rho C_2 \varepsilon^2}{\mathrm{k} + \sqrt{\nu\varepsilon}} + \nabla\cdot((\mu + \mu_t / \sigma_\varepsilon)\nabla \varepsilon) \tag{5}$$

Where ρ, $\boldsymbol{U}$, ∇, p, μ, μ_t, μ_{eff}, Δ, c_{p}, T, λ, S_{T}, k, P_{k}, ε, σ_k, σ_ε, C_1, C_2 denoted density, velocity vector, gradient operator, pressure, dynamic viscosity, turbulence viscosity, effective dynamic viscosity, Laplace operator, specific heat, temperature, thermal conductivity, heat source term, turbulent kinetic energy, generation term, turbulent dissipation rate, turbulent Prandtl numbers corresponding to turbulent kinetic energy and turbulent dissipation rate, and constants. For copper plate-fin heat sinks, a porous medium model was adopted to alleviate computational overhead, whose feasibility was validated by Kim et al. [38]. The pressure drop across the finned structures was characterized as:

$$\frac{F_d}{0.5\rho u_{\mathrm{ch}}^2} = f_{\mathrm{app}} N(2HL + w_{\mathrm{ch}} L) + K_c(HW) + K_e(HW) \tag{6}$$

$$u_{\mathrm{ch}} = u_{\mathrm{f}}(1 + t / w_{\mathrm{ch}}) \tag{7}$$

$$f_{\mathrm{app}}\mathrm{Re}_D = \left[\left(\frac{3.44}{\sqrt{L^*}}\right)^2 + (f\mathrm{Re}_D)^2\right]^{0.5} \tag{8}$$

$$L^* = L / (D_h \mathrm{Re}_D) \tag{9}$$

$$f\mathrm{Re}_D = 24 - 32.527\left(\frac{w_{\mathrm{ch}}}{H}\right) + 46.721\left(\frac{w_{\mathrm{ch}}}{H}\right)^2 - 40.829\left(\frac{w_{\mathrm{ch}}}{H}\right)^3 + 22.954\left(\frac{w_{\mathrm{ch}}}{H}\right)^4 - 6.089\left(\frac{w_{\mathrm{ch}}}{H}\right)^5 \tag{10}$$

$$K_c = 0.42(1 - \sigma^2) \tag{11}$$

$$K_e = (1 - \sigma^2)^2 \tag{12}$$

$$\sigma = 1 - N\cdot t / W \tag{13}$$

where H, W, L, f_{app}, u_{ch}, u_{f}, t, w_{ch}, L^*, D_{h}, Re_{D}, f, K_{c}, K_{e}, σ were fin height, heat sink width, fin length, apparent friction factor, channel velocity, free stream velocity, fin thickness, fin space, normalized fin length, hydraulic diameter, Reynolds number, friction factor, contraction loss coefficient, expansion loss coefficient, and porosity. The permeability $1/\alpha$ and inertial resistance coefficients C required for the additional momentum sink term can be calculated as

$$\Delta p = c_1 u_i + c_2 u_i^2 \tag{14}$$

$$\frac{\Delta p}{\Delta x_i} = S_i = \frac{\mu}{\alpha} u_i + \frac{1}{2}\rho C u_i^2 \tag{15}$$

$$\frac{1}{\alpha} = \frac{c_1}{\mu \Delta x_i} \tag{16}$$

$$C = \frac{2c_2}{\rho \Delta x_i} \tag{17}$$

Given that the rectangular RAM modules contribute negligible thermal load to the system, they were also modeled as porous media to optimize computational efficiency. Pressure-velocity coupling was achieved using the coupled algorithm. Spatial discretization of the momentum and energy equations employed a second-order upwind scheme, while the turbulent kinetic energy and dissipation rate equations were resolved using a first-order upwind scheme to enhance stability. Convergence was deemed achieved when the normalized residuals fell below 10^{-3} for continuity, 10^{-5} for momentum and

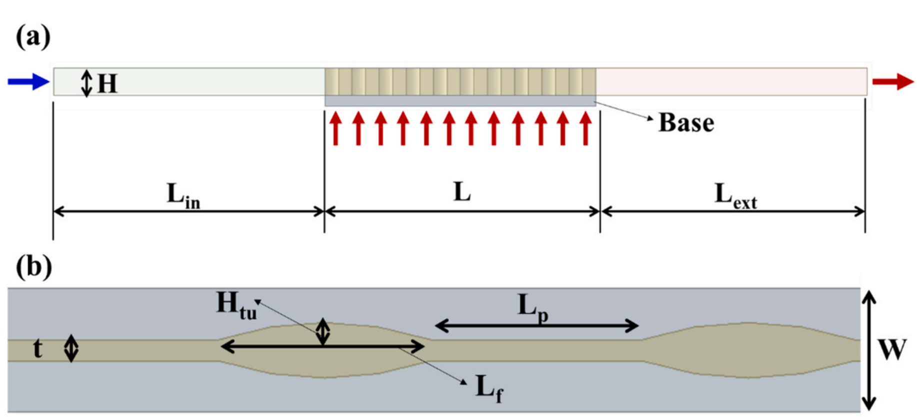

Fig. 3. (a) Frontal view of an individual rib unit structure in heat sink design. (b) Schematic diagram of periodic architecture in a single fin structure.

turbulence, and 10^{-7} for energy.

A rigorous grid independence study was conducted to ensure that the numerical results were mesh-insensitive. As illustrated in Fig. S1, a mesh density of approximately 1.6 million cells was found to be sufficient, with further refinement resulting in temperature variations of less than 0.1 °C. Utilizing this optimized mesh and FC-40 as the working fluid, the model was validated against experimental data from Ref. [21]. Four turbulence models—Standard k-ε, RNG k-ε, Realizable k-ε, and SST k-ω—were evaluated at a flow rate of 3 LPM and a heat load of 300 W. As shown in Fig. 2(a), while all models exhibited reasonable agreement with experimental values, the Realizable k-ε model demonstrated superior numerical robustness and was therefore selected for all subsequent server and tank-level simulations. Furthermore, Fig. 2(b) confirmed that the CFD results aligned closely with experimental data across varying power densities, with a maximum deviation of less than 5%.

Fig. 3(a) presented the frontal view of an individual rib unit within the heat sink, where H, L_{in}, L, and L_{ext} denoted the fin height, inlet section length, design section length, and extension section length, respectively. The heat sink was fabricated from copper with a base thickness of 2 mm. The periodic arrangement of the fin elements was illustrated in Fig. 3(b), with parameters t, W, H_{tu}, L_{f}, and L_{p} representing the fin thickness, overall width, protruding height, arc-shaped feature length, and inter-unit spacing. In this study, a uniform periodic design was adopted where L_{f} and L_{p} were maintained equal throughout the structure. These arc-shaped fins were specifically designed to enhance convective heat transfer by promoting flow reattachment and disrupting boundary layer development. Detailed geometric dimensions were listed in Table S1 and a three-dimensional view of an individual fin was provided in Fig. S2.

2.2. Machine learning and optimization methods

Directly coupling Computational Fluid Dynamics (CFD) simulations with evolutionary algorithms entailed a prohibitive computational cost, particularly when navigating high-dimensional design spaces. To circumvent this, a systematic parametric study was first conducted to characterize the influence of the heat sink's geometric features on its thermohydraulic performance. The primary design variables, specifically H_{tu}, L_{f} and L_{p}, were non-dimensionalized using the Buckingham π theorem to establish a robust set of governing dimensionless groups. Consequently, the Reynolds number (Re), Nusselt number (Nu), friction factor (f), and performance evaluation criterion (PEC) were defined as follows:

$$Re = \rho u D_h / \mu \tag{18}$$

$$Nu = q_w \big/ \left(T_w - T_f\right) \tag{19}$$

$$f = \frac{2\Delta p D_h}{\rho u^2 L} \tag{20}$$

$$\text{PEC} = \frac{(Nu/Nu_0)}{(f/f_0)^{1/3}} \tag{21}$$

where q_w, T_w, and T_f denoted the heat flux, average wall temperature, and average fluid temperature, while Nu_0 and f_0 represented the baseline Nusselt number and friction factor for a straight-fin heat sink of equivalent dimensions. The dimensional analysis was performed by selecting D_{h} as the characteristic length scale, which guaranteed geometric similarity through normalization of all length-based parameters:

$$\pi_1 = \frac{H_{tu}}{D_h}, \pi_2 = \frac{t}{D_h}, \pi_3 = \frac{H}{D_h}, \pi_4 = \frac{L}{D_h}, \pi_5 = \frac{L_p}{D_h}, \pi_6 = \frac{L_f}{D_h}, \pi_7 = \frac{L}{(L_p + L_f)} \tag{22}$$

The final dimensionless parameter π_7, representing the count of periodic units, was truncated to the nearest integer using the floor function, ensuring consistency with the discrete nature of the physical system.

The dimensionless parameter π_7, representing the number of periodic units, was truncated to the nearest integer using the floor function to maintain consistency with the discrete physical constraints of the system. To predict performance across the design space, empirical correlations for Nu and f were developed in the following form:

$$Nu = Re^a \pi_1^b \pi_2^c \pi_3^d \pi_4^e \pi_5^f \pi_6^g \pi_7^h \tag{23}$$

$$f = Re^{a2} \pi_1^{b2} \pi_2^{c2} \pi_3^{d2} \pi_4^{e2} \pi_5^{f2} \pi_6^{g2} \pi_7^{h2} \tag{24}$$

where a, b, c, d, e, f, g, h were fitted coefficients. To facilitate parameter estimation, logarithmic transformation was applied to both sides of the correlation. To improve accuracy, the data-driven surrogate model was also constructed using Gaussian Process Regression (GPR) with hyperparameter tuning via Bayesian optimization. This physics-informed proxy model was then coupled with an evolutionary algorithm to perform optimization of the heat sink geometry, targeting maximal PEC. GPR models were nonparametric kernel-based probabilistic models. A GP was a set of random variables, such that any finite number of them had a joint Gaussian distribution. A GP was defined by its mean function $m(x)$ and covariance function $cov\ (x,x')$, which can be written as.

Where a through h were fitted coefficients. To linearize the parameter estimation process, a logarithmic transformation was applied to both sides of the equations. To further enhance predictive fidelity, a data-driven surrogate model was constructed using Gaussian Process Regression (GPR), with hyperparameters fine-tuned via Bayesian optimization. This physics-informed proxy model served as a high-fidelity surrogate, which was subsequently coupled with an evolutionary algorithm framework to optimize the geometry for maximal PEC. As a nonparametric, kernel-based probabilistic approach, a GP was defined by a collection of random variables, any finite number of which followed a joint Gaussian distribution. The GP was fully characterized by its mean function $m(x)$ and covariance function $cov\ (x,x')$, expressed as:

$$E(f(x)) = m(x) \tag{25}$$

$$cov(x, x') = E[[f(x) - m(x)][f(x') - m(x')]] \tag{26}$$

Evolutionary algorithms encompassed a family of stochastic optimization algorithms inspired by biological evolution and collective intelligence, including but not limited to genetic algorithms (GAs), and particle swarm optimization (PSO). Ge et al. [39] and Liu et al. [40,41] employed GAs to optimize the microchannel heat sinks and the thermal-hydraulic performance of circular tubes, respectively. Therefore, we primarily employed a GA to optimize the geometric parameters of heat sinks, while utilizing PSO for cross-validation. All optimization procedures were implemented in MATLAB to ensure computational efficiency and reproducibility. For multi-objective scenarios, the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) was integrated to select the most favorable compromise solution from the Pareto front. The operational steps for the TOPSIS-based screening are as follows.

(1) Establish the evaluation matrix and calculate the normalized matrix;
(2) Determine the positive and negative ideal alternatives;
(3) Compute the distance from the Pareto solution to the positive and negative ideal alternative;
(4) Compute the relative closeness to the ideal solution for each Pareto solution;
(5) Rank the alternatives and choose the compromise solution.

2.3. The experimental setup

Fig. 4(a) and (b) illustrated the schematic diagram and physical

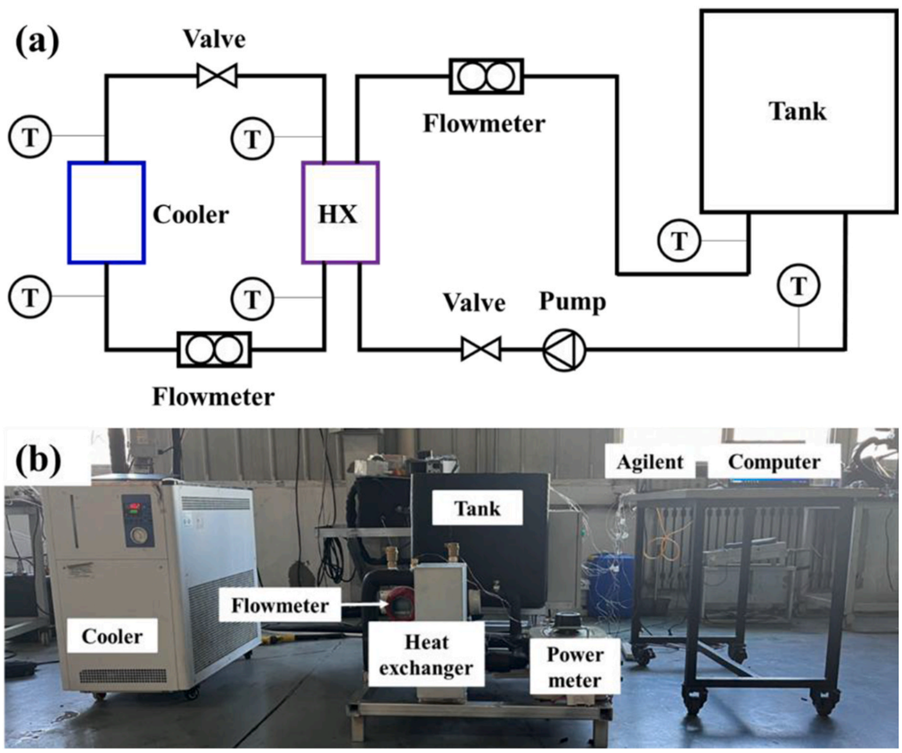

Fig. 4. (a) Schematic and (b) physical layout of the single-phase immersion cooling experimental setup.

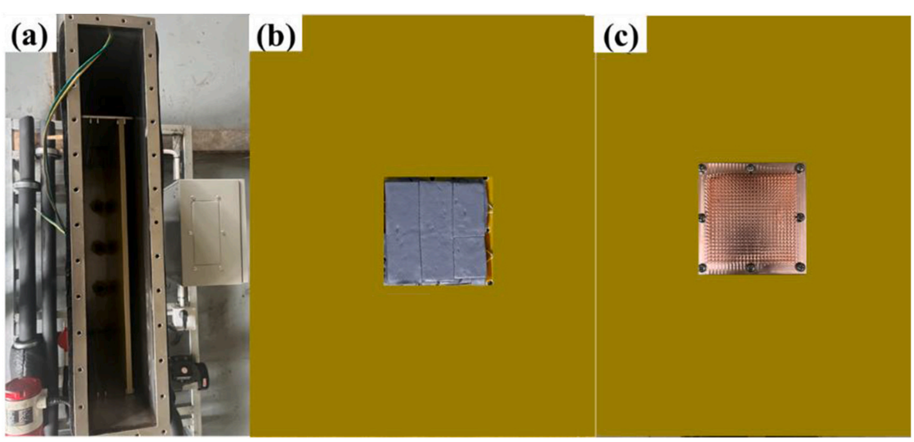

Fig. 5. The physical map of (a) immersion tank, (b) thermal pad, ceramic heater, epoxy resin board, (c) and copper heat sink.

layout of the SPIC experimental system, respectively. The system consisted of several key components: an immersion tank, coolant and water flow meters, a coolant pump, a power regulator, a plate heat exchanger, a chiller, an Agilent data acquisition system, and a laptop for data processing.

Fig. 5(a) displayed a physical image of the stainless steel immersion tank utilized in the experiment. The internal dimensions were 61 cm × 50 cm × 10 cm and the wall thickness was 5 mm. The positioning plate was mounted on the overflow plate, maintaining a precisely controlled 2 cm axial separation from the stainless steel plate on the back. The ceramic heater served as the simulated heat source, whose main ingredient was alumina (Al_2O_3). The thermal pad was used to stick the heating element to an epoxy resin board, as shown in Fig. 5(b). The dimensions of ceramic heater and epoxy resin board were 50 mm × 50 mm × 2 mm and 49.7 cm × 40 cm × 1 cm. The ceramic heating element was centrally positioned on the epoxy resin substrate, with one corner of the heating element 13.5 cm away from the top and 18.35 cm away from the right or left edge. To minimize thermal contact resistance and improve measurement fidelity, the thermal pad was applied between the ceramic heater and the copper interposers. The T-type thermocouple was inserted into the hole of the copper interposer and bonded to them. Based on the heat transfer correlation and experimental test, we found that the heat transfer coefficient of a smooth heating plate was inherently limited. Hence a needle-fin heat sink was integrated on the heat source to enhance heat transfer, as shown in Fig. 5(c). The fin array was strategically localized to the central 100 × 100 mm^2 region to balance

Table 1
Physical properties of the working fluid.

Physical property	Silicone oil	SS110
Density (kg/m^3)	930	1738
Thermal conductivity (W/(m·K))	0.15	0.063
Specific heat (J/(kg·K))	1850	1055
Dynamic viscosity (Pa·s)	9.3×10^{-3}	1.18×10^{-3}

Table 2
Fluid and thermo-physical properties of different SPIC coolants.

Name	Density ρ (kg/m^3)	Specific heat c_p (J/(kg·K))	Thermal conductivity λ (W/(m·K))	Dynamic viscosity μ (Pa·s)	Pr
FC-40	1820.46	1076	0.0642	0.00273	45.755
FC-3283	1781.78	1076	0.0628	0.00137	23.47
Noah3000A	1775	1169	0.0611	0.00174	33.2907
HT135	1691.62	960	0.065	0.00169	24.96
SF10	1580	1240	0.077	0.0071	114.34
TMC-40	1856	1100	0.065	0.0041	69.38
SS110	1738	1055	0.065	1.18e-3	19.76
EC-110	832.7	2209	0.136	0.00536	87.06
EC-120	820	2211.9	0.1439	0.0132	202.9
EC-122	810	2210	0.1359	0.00407	66.19
EC-130	830	2203	0.1467	0.02996	449.91
EC-140	840	2191.2	0.1568	0.05636	787.60
PAO-6	811.1	2218	0.1546	0.025	358.67
PAO-4	819	2042.1	0.153	0.0176	234.91
DC-550	934	1500	0.13	0.07846	905.31
S5 X	806	2274	0.142	0.0079	126.51
Crystal Plus 70T	825	2220	0.14	0.0123	195.04

Table 3
Comparison of natural convection heat transfer for typical SPIC coolants.

Coolant	*Gr*	*Ra*	*Nu*	h_n (W/(m^2·K))	*Q* (W)
STE mineral oil	226.13	30384.73	7.79	202.53	20.25
PAO-6 oil	36.09	12945.57	6.29	194.59	19.46
EC-110	935.3	81427.5	9.97	271.09	27.11
Silicone oil	2033.94	120063.6	10.98	263.58	26.36
FC-40	26146.55	1196465.91	19.51	250.55	25.05
HT135	54003.38	1347924.48	20.10	261.34	26.13
SS110	148356.4	2931522.49	24.41	307.61	30.76

thermal performance with manufacturing efficiency. The dimension of single fin was 2 mm × 2 mm × 25 mm and the fin pitch was 2 mm.

The thermophysical properties of the SS110 coolant and silicone oil were summarized in Table 1. Notably, the dynamic viscosity of SS110 was significantly lower than that of silicone oil. This characteristic was expected to yield a substantially thinner boundary layer thickness (BLT), thereby facilitating superior flow mobility and enhanced convective heat transfer efficiency in the immersion cooling system.

3. Results and discussion

3.1. Thermohydraulic performance evaluation of various SPIC coolants

3.1.1. The results of the theoretical analysis

A systematic collection of thermophysical properties was conducted for various SPIC coolants, encompassing mineral oil, silicone oil, and fluorocarbon-based liquids. As presented in Table 2, key parameters include density, specific heat, thermal conductivity, dynamic viscosity, and Prandtl number (Pr).

Theoretical analysis based on natural convection principles was conducted to investigate the thermal performance limits of SPIC systems. Assuming a chip surface area of 25 cm^2 and a temperature difference of 40 K, the dimensionless Grashof number (*Gr*) range was calculated to be smaller than 3×10^9 using the correlation $Gr = g\beta\Delta TL^3/\nu^2$, where g, β, ΔT, L, and ν were gravitational acceleration, thermal expansion coefficient, temperature difference, characteristic length, and kinematic viscosity. As shown in Table 3, the heat transfer coefficients estimated by empirical correlations were below 100 W/(m^2·K), which were insufficient for effective thermal management of modern high-performance servers. Consequently, forced convection strategies emerged as essential technologies to enhance heat dissipation.

Fig. 6 presented a radial comparison of the thermophysical properties of mineral oil, silicone oil, and fluorocarbon coolants via a polar diagram. Oil-based coolants exhibited kinematic viscosities one to two orders of magnitude higher than their fluorocarbon counterparts. This pronounced viscosity contrast was fundamentally rooted in their molecular architectures. Oil-based coolants comprised long-chain hydrocarbons that induced significant intermolecular friction, thereby compromising flowability. Conversely, fluorocarbon coolants showed approximately 50% lower thermal conductivity and specific heat capacity compared to oils. These limitations were attributed to the compact molecular structure and strong C-F bonds of fluorocarbons, which restricted vibrational energy transfer. However, fluorocarbon coolants exhibited 60–80% higher density, necessitating structural reinforcements in immersion tank designs. Despite these drawbacks, their superior dielectric properties and chemical stability offered

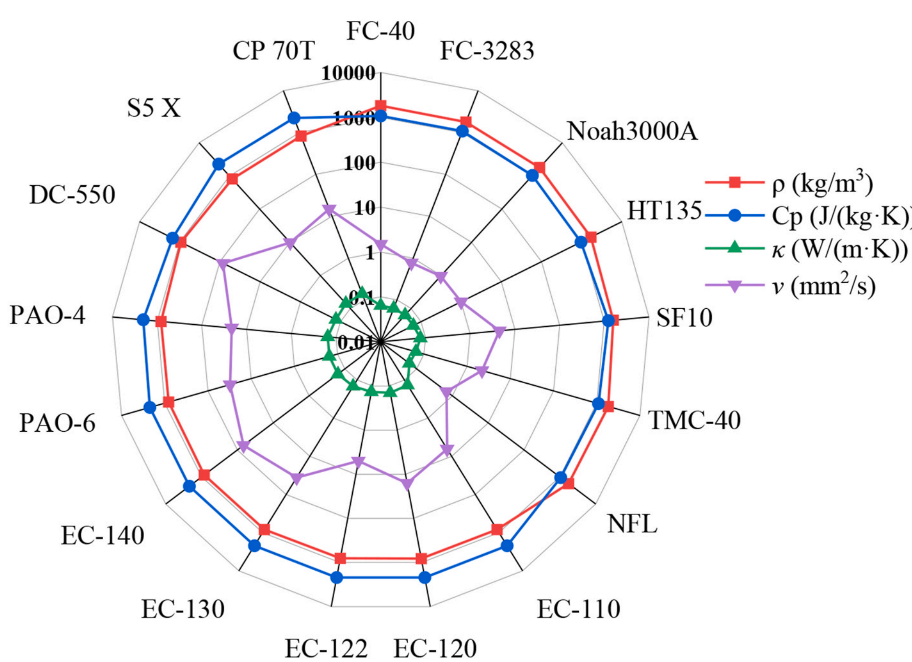

Fig. 6. Thermophysical properties comparison of mineral oil, silicone oil, and fluorocarbon coolants.

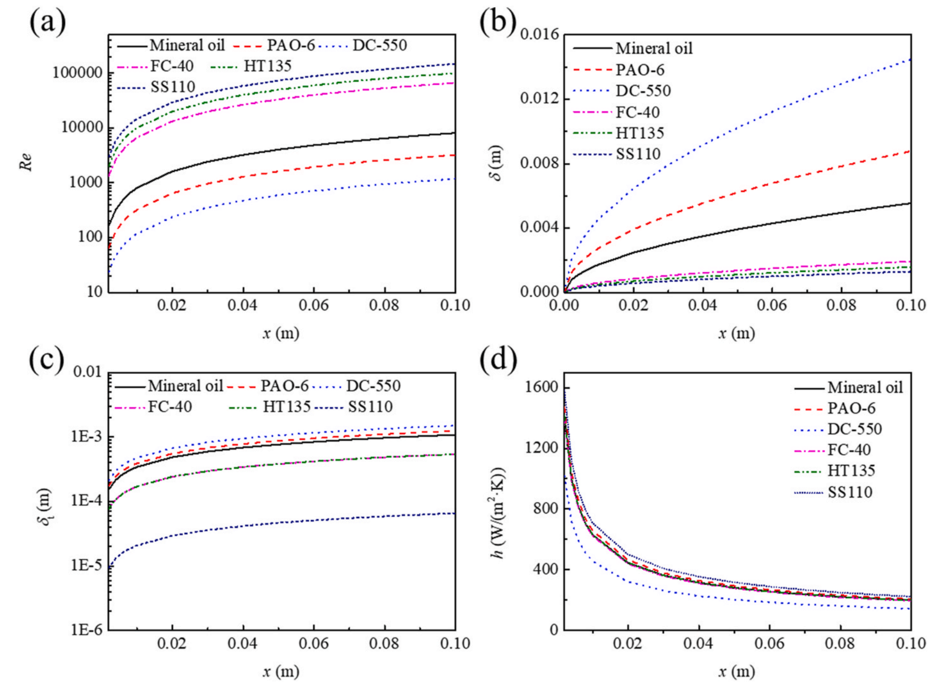

Fig. 7. Variation of key parameters for typical SPIC coolants flowing over a flat plate at a flow velocity of 1 m/s: (a) Reynolds number; (b) velocity boundary layer thickness; (c) thermal boundary layer thickness; (d) convective heat transfer coefficient.

compensatory advantages, emphasizing the need to select SPIC coolants based on specific application requirements.

To further elucidate the flow and heat transfer characteristics, Reynolds numbers were calculated for external flow over an isothermal plate at a velocity of 1 m/s, utilizing the properties listed in Table 3 (STE mineral oil, PAO-6, DC-550, FC-40, HT135, and SS110). As illustrated in Fig. 7(a), all Re values remained below the critical threshold of 5 × 1055 × 105, confirming a laminar flow regime. The velocity boundary layer thickness (δ) and the thermal boundary layer thickness (δ_t) were calculated as follows:

$$\delta = 5x \Big/ \sqrt{Re} \tag{27}$$

$$\delta_t \approx \delta / Pr^{1/3} \tag{28}$$

Fig. 7(b) and (c) compared the velocity and thermal boundary layer thicknesses of STE mineral oil, PAO-6 oil, DC-550 silicone oil, FC-40, HT135, and SS110. The velocity boundary layer thicknesses were ranked in the following order: DC-550 silicone oil > PAO-6 oil > mineral oil > FC-40 > HT135 > SS110. The thermal boundary layer thicknesses followed the same ranking. The heat flux removed by the coolant was:

$$q = -\kappa \frac{\partial T}{\partial y}\Big|_{y=0} \approx \kappa (T_s - T_\infty) \Big/ \delta_t \tag{29}$$

Consequently, for a single-phase immersion coolant, a higher thermal conductivity, combined with a thinner thermal boundary layer, resulted in greater heat removal. The local surface convective heat transfer coefficient can be expressed as:

$$h = 0.332\, \kappa / x Re^{0.5}\, Pr^{1/3} \tag{30}$$

Fig. 7(d) illustrated the ranking of the local convective heat transfer coefficients for the coolants as follows: SS110 > HT135 > FC-40 > mineral oil > PAO-6 oil > DC-550 silicone oil.

For laminar flow over a heated surface, the convective heat transfer coefficient scales with $Re^{0.5}Pr^{1/3}$. At a constant velocity, this coefficient was governed by the thermophysical property group $\rho^2 c_p^2 k^4/\nu$. Although oil-based coolants offered comparable volumetric heat capacity and nearly doubled the thermal conductivity of fluorocarbons, these benefits were often negated by their substantially higher kinematic viscosity. Specifically, if the kinematic viscosity of an oil-based coolant exceeded that of a fluorocarbon by more than eightfold, its convective heat transfer performance became inferior. This relationship is further explored through entransy dissipation within the laminar thermal boundary layer, given by

$$J_d = \int_0^{\delta_t} \kappa_f \left(\frac{\partial T}{\partial y}\right)^2 dy = q_{wall}^2 \Big/ h \approx \frac{q_{wall}^2 \delta_t}{\kappa_f} \tag{31}$$

Therefore, reducing δt or increasing the coolant's thermal conductivity directly mitigated entransy dissipation, enhancing thermal efficiency.

In the turbulent regime, the convective heat transfer coefficient scaled with $Re^{0.8}Pr^{1/3}k$. For a given velocity, this coefficient was governed by the factor $\rho^5 c_p^5 k^{10}/\nu^7$. If the kinematic viscosity of an oil-based coolant was more than 2.7 times that of a fluorocarbon, its heat transfer performance fell behind. Furthermore, in regimes where fluorocarbons transitioned to turbulence while oils remained laminar, the former exhibited a significantly higher heat transfer coefficient due to the turbulent viscous sublayer being substantially thinner than the laminar boundary layer. The entransy dissipation within the turbulent thermal boundary layer can be expressed as:

$$J_{d,turb} = \int_0^{\delta_t} \kappa_f \left(\frac{\partial T}{\partial y}\right)^2 dy = q_{wall}^2 \Big/ h_{turb} \approx \frac{q_{wall}^2 \delta_{vs}}{\kappa_f} \tag{32}$$

where h_{turb} was the turbulent convective heat transfer coefficient and δ_{vs} was the thickness of the viscous sublayer.

For the case of a SPIC coolant flowing over a heated plate, integrating the energy equation for convective heat transfer within the thermal boundary layer yielded:

$$\rho c_p \int_0^{\delta_t} |\boldsymbol{U}| \times |\nabla T| \cos\beta dy = -\kappa_f \left(\frac{\partial T}{\partial y}\right)\Big|_{y=0} \tag{33}$$

Here, β was the angle between the velocity field and the temperature gradient field, which can be defined as:

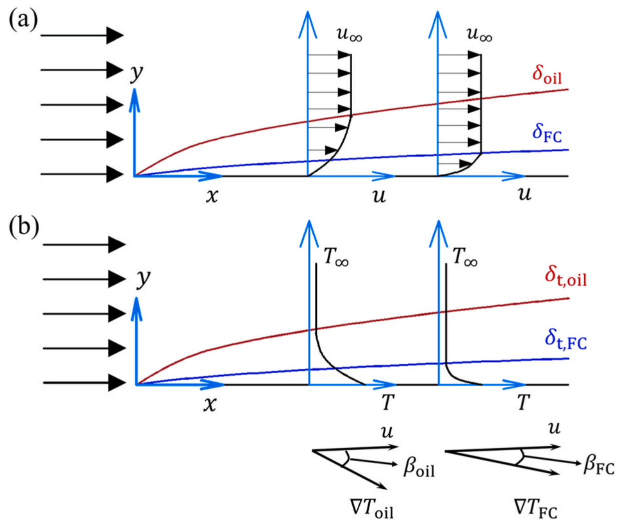

Fig. 8. Schematic of the boundary layers for SPIC coolants flowing over a heated plate: (a) Velocity boundary layer; (b) Thermal boundary layer.

$$\cos \beta = (\boldsymbol{U} \cdot \nabla T)/(|\boldsymbol{U}| \times |\nabla T|) \tag{34}$$

Therefore, a smaller synergy angle, β, was more desirable. Fig. 8(a) and (b) were schematics of the velocity and thermal boundary layers for oil-based and fluorocarbon coolants at an identical velocity. The thermal boundary layer of a fluorocarbon coolant was thinner than that of an oil-based coolant, which led to a smaller angle between the velocity and temperature gradient fields. Consequently, the synergy between velocity and the temperature gradient was enhanced when using a fluorocarbon coolant.

For the more general case of a server with constant heat generation, neglecting viscous dissipation and internal heat sources, the energy equation for steady-state convective heat transfer was given by:

$$\rho c_p \boldsymbol{U} \cdot \nabla T = \nabla \cdot (\kappa_f \nabla T) \tag{35}$$

Expressing the left-hand side of equation (35) in terms of the synergy angle yielded:

$$\rho c_p |\boldsymbol{U}||\nabla T| \cos \beta = \nabla \cdot (\kappa_f \nabla T) \tag{36}$$

Multiplying both sides of equation (36) by the temperature T and then rearranging yielded:

$$\rho c_p T |\boldsymbol{U}||\nabla T| \cos \beta = \nabla \cdot (\kappa_f T \nabla T) - \kappa_f |\nabla T|^2 \tag{37}$$

The last term of equation (37) was precisely the entransy dissipation, and can be further rearranged to yield:

$$\kappa_f |\nabla T|^2 = \nabla \cdot (\boldsymbol{q} T) - \rho c_p T |\boldsymbol{U}||\nabla T| \cos \beta \tag{38}$$

where $\boldsymbol{q}$ was conductive heat flux. Therefore, reducing the synergy angle also reduced entransy dissipation.

For high-power chips, finned heat sinks were utilized to increase the heat transfer area and reduce the chip temperature. As shown in Fig. 9, when a low-temperature, SPIC coolant flowed over the finned heat sink, the small fin spacing resulted in high flow resistance, causing the coolant flow to split. A portion of the coolant flowed through the fin gaps at a lower velocity to exchange heat with the fins, while another portion bypassed the main body of the heat sink, flowing out from its sides, which was detrimental to heat dissipation.

For a typical plate-fin heat sink, the resulting pressure drop [42] was given by:

$$\Delta p = 0.5 \rho U^2 (f_c + f_e + 4fL / D_h) \tag{39}$$

where f_c, f_e, and f were the contraction coefficient, expansion coefficient, and friction factor. L and D_h were the length of the heat sink in the flow direction and the hydraulic diameter. In single-phase immersion liquid cooling, the frictional pressure drop within the channel was dominant. Consequently, the friction factor was inversely proportional to the Reynolds number and can be written as:

$$f \propto 1/Re \tag{40}$$

The pumping power, defined as the product of the flow rate and pressure drop, was therefore proportional to the product of dynamic viscosity and velocity squared, which can be written as:

$$P_{pump} \propto \mu U^2 \tag{41}$$

When the pump power consumption was identical, the flow velocity was inversely proportional to the 0.5th power of the dynamic viscosity, which can be expressed as:

$$U \propto \mu^{-0.5} \tag{42}$$

Therefore, for the same pumping power, fluorocarbon coolants with lower dynamic viscosity achieved a higher flow velocity than oil-based coolants. For a more general case, where the friction factor was inversely proportional to the Reynolds number raised to the power of α, the pumping power was related to the dynamic viscosity, density, and velocity, and can be written as:

$$P_{pump} \propto \mu^{\alpha} \rho^{1-\alpha} U^{3-\alpha} \tag{43}$$

When the pump power consumption was constant, the flow velocity was correlated with density and dynamic viscosity, which can be expressed as:

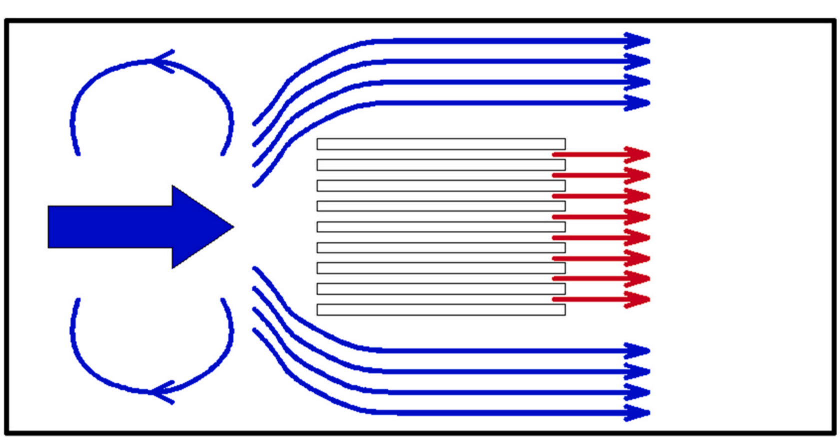

Fig. 9. Schematic of a SPIC coolant flowing over a finned heat sink on a chip.

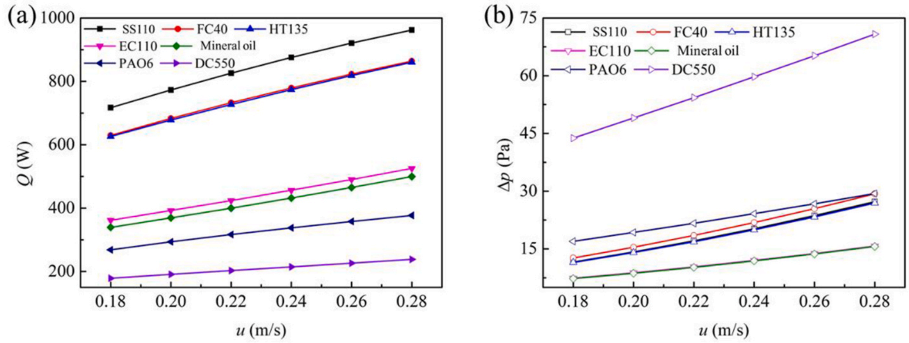

Fig. 10. The thermohydraulic characteristics of different single-phase immersion coolants: (a) Heat transfer rate; (b) Pressure drop.

$$U \propto \left(\mu^{-\alpha}\rho^{\alpha-1}\right)^{\frac{1}{3-\alpha}} = \left(\nu^{-\alpha}\rho^{-1}\right)^{\frac{1}{3-\alpha}} \tag{44}$$

As α was between 0 and 1, the flow velocity of a SPIC coolant at a constant pumping power was inversely proportional to the product of its density and kinematic viscosity raised to the power of α.

When the same heat sink was used, the conductive thermal resistance was identical. Therefore, for different SPIC coolants, the distinction lied in the convective heat transfer process. For a chip with a heat generation of Q, the average outlet temperature of the coolant from the heat sink was given by:

$$T_{\mathrm{out}} = T_{\mathrm{in}} + Q / \left(\rho q_{\mathrm{v}} c_{\mathrm{p}}\right) \tag{45}$$

The relationship between the heat current Q and the average temperature of the heat sink base T_{w} can expressed as:

$$Q = KA(T_{\mathrm{w}} - 0.5(T_{\mathrm{in}} + T_{\mathrm{out}})) \tag{46}$$

where K denoted the average heat transfer coefficient. The average temperature of the heat sink base can be expressed as:

$$T_{\mathrm{w}} = T_{\mathrm{in}} + Q / KA + Q / \left(2\rho q_{\mathrm{v}} c_{\mathrm{p}}\right) \tag{47}$$

Therefore, for identical heat transfer rates and inlet temperatures, the average heat sink temperature depended on $1/(KA)+1/(2\rho q_{\mathrm{v}} c_{\mathrm{p}})$. This term corresponded to the thermal resistance, which employed the temperature difference between the heat sink base and the inlet as the driving force for heat transfer. For a plate-fin heat sink, the specific expression for this thermal resistance [43] was given by:

$$R_{\mathrm{s,in}} = \frac{t}{\kappa_{\mathrm{s}} LW} + \frac{(w_{\mathrm{c}} + w_{\mathrm{w}})H}{3\kappa_{\mathrm{s}} LW w_{\mathrm{w}}} + \frac{17}{140}\frac{(w_{\mathrm{c}} + w_{\mathrm{w}})w_{\mathrm{c}}}{\kappa_{\mathrm{f}} LWH} + \frac{\sqrt{12\mu_{\mathrm{f}}(w_{\mathrm{c}} + w_{\mathrm{w}})L}}{\rho c_{\mathrm{p}}\sqrt{WHP_{\mathrm{pump}} w_{\mathrm{c}}^{3}}} \tag{48}$$

where t, L, W, H, w_{c}, and w_{w} denoted the thickness, length, and width of the heat sink base, fin height, fin pitch and fin thickness. The terms κ_{s}, κ_{f}, ρ, and c_{p} were the thermal conductivities of the solid and coolant, density, and specific heat capacity, while P_{pump} was the pumping power. For a fixed heat sink geometry, the conductive thermal resistance remained constant. Given that the volumetric heat capacities of various SPIC coolants exhibited minimal variation, and the thermal conductivity of oil-based coolants was approximately twice that of fluorocarbon coolants, the condition for fluorocarbon coolants to outperform oil-based coolants under identical pumping power was derived as follows:

$$\sqrt{\mu_{\mathrm{oil}}} - \sqrt{\mu_{\mathrm{FC}}} < \frac{17}{560\sqrt{3}}\frac{\rho c_{\mathrm{p}}\sqrt{P_{\mathrm{pump}}(w_{\mathrm{c}} + w_{\mathrm{w}})w_{\mathrm{c}}^{5}}}{\kappa_{\mathrm{FC}}\sqrt{HWL^{1.5}}} \tag{49}$$

While the arrangement of staggered fins enhanced fluid mixing, it concurrently led to greater pumping power requirements, a lower proportion of high-conductivity solid material, elevated conductive thermal resistance, and a reduced heat exchange surface. Consequently, a staggered-fin heat sink may not exhibit a lower overall thermal resistance compared to its plate-fin counterpart. A detailed analysis of the flow and heat transfer within the fin pitch was presented next. In the case of laminar convection, the convective heat transfer coefficient was determined by the Sieder-Tate correlation, which can be expressed as:

$$Nu = 1.86\left(\frac{RePr}{L/D_{\mathrm{h}}}\right)^{\frac{1}{3}}\left(\mu_{\mathrm{f}}/\mu_{\mathrm{w}}\right)^{0.14} \tag{50}$$

The hydraulic diameter D_{h} can be expressed in terms of the cross-sectional flow area A_{cs}, the flow direction length L, and the heat transfer area A, which was given by:

$$D_{\mathrm{h}} = 4A_{\mathrm{cs}}L/A \tag{51}$$

The thermal resistance within a single channel was expressed as:

$$R_{\mathrm{channel}} = 1 \Bigg/ \left[1.86\left(\rho c_{\mathrm{p}} U \kappa_{\mathrm{f}}^{2}\right)^{\frac{1}{3}}\left(\frac{\mu_{\mathrm{f}}}{\mu_{\mathrm{w}}}\right)^{0.14}\left(\frac{A^{4}}{4A_{\mathrm{cs}}L^{2}}\right)^{\frac{1}{3}}\right] + \frac{1}{2\rho c_{\mathrm{p}} U A_{\mathrm{cs}}} \tag{52}$$

Therefore, the convective heat transfer performance of different coolants was related to their thermophysical properties, flow velocity, and geometric dimensions. This implied that optimal thermal performance, characterized by low thermal resistance, ariseed from combining coolants with high volumetric specific heat capacity, high thermal conductivity, and low viscosity with heat sinks that provided a large heat transfer area. Equation (52) further suggested a design principle: high-viscosity oil-based coolants should be used with wider fin spacing to maintain sufficient flow velocity, whereas low-viscosity fluorocarbon-based coolants can be employed with narrower fin pitch to exploit a larger heat transfer area. For narrow fin spacing, the flow quickly became fully developed, and under constant wall temperature or heat flux boundary conditions, the Nusselt number remained constant ($Nu = \Gamma$). The total thermal resistance, R_{channel}, was therefore given by:

$$R_{\mathrm{channel}} = \frac{D_{\mathrm{h}}}{\Gamma\kappa_{\mathrm{f}} A} + \frac{1}{2\rho c_{\mathrm{p}} U A_{\mathrm{cs}}} \tag{53}$$

Based on the characteristics of the thermophysical parameters of SPIC coolants, the condition for the thermal resistance of the former to be higher than that of the latter was derived as:

$$\frac{1}{U_{\mathrm{oil}}} - \frac{1}{U_{\mathrm{FC}}} > 4\rho c_{\mathrm{p}} L \Big/ (\Gamma\kappa_{\mathrm{FC}})(A_{\mathrm{cs}}/A)^{2} \tag{54}$$

For a given pumping power, fluorocarbon coolants demonstrated better thermo-hydraulic performance than oil-based coolants, especially in configurations with small fin spacing, consistent with the analyzed effects of fluid properties.

3.1.2. The results of the numerical simulation

To investigate the heat transfer and pressure drop characteristics of 7 SPIC coolants under steady-state conditions, CFD simulations were conducted on a 1U server model shown in Fig. 1. The coolants evaluated included fluorocarbon-based (SS110, FC-40, HT-135), EC-110 synthetic

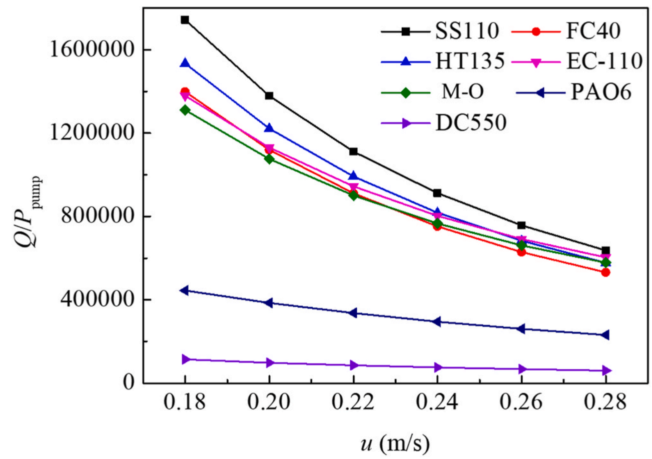

Fig. 11. Heat removal per unit pumping power for single-phase immersion coolants.

oil, PAO-6 oil, and DC-550 silicone oil. A high-resolution mesh of 2.08 million elements was employed to ensure numerical accuracy, with boundary conditions set to a chip surface temperature of 80 °C and a coolant inlet temperature of 35 °C. The resulting heat removal rates and pressure drops across various flow velocities were depicted in Fig. 10(a) and (b), respectively.

Ferrouillat et al.'s energy performance criterion [42], defined as $Q/P_{\text{pump}} = Q/(q_v \cdot \Delta p)$, was adopted to quantify the comprehensive thermo-hydraulic performance of these coolants. Fig. 11 illustrated the heat removal per unit pumping power as a function of flow velocity for various SPIC coolants. In summary, while both the heat transfer rate and pressure drop increased with flow velocity, the heat removal per unit pumping power exhibited a declining trend. This indicated that the rate of increase in pumping power surpassed that of the heat transfer rate. The heat transfer coefficients of fluorocarbon-based coolants were 80%–100% higher than those of oil-based coolants. With the exception of DC-550, the pressure drops for the other coolants were comparable. The pressure drop for DC-550 was approximately three times greater than that of the other coolants, a discrepancy attributable to its significantly higher viscosity ($\nu = 84$ mm^2/s).

To investigate the underlying reasons for performance disparities among various SPIC coolants, a detailed model of a finned heat sink was developed, incorporating a mesh of 5.62 million cells. CFD simulations were then performed for 5 coolants (SS110, FC40, EC110, PAO6, and DC550) at a heating power of 300 W. Furthermore, entransy dissipation was calculated to quantify the irreversible losses associated with temperature difference, which was calculated using the following equation:

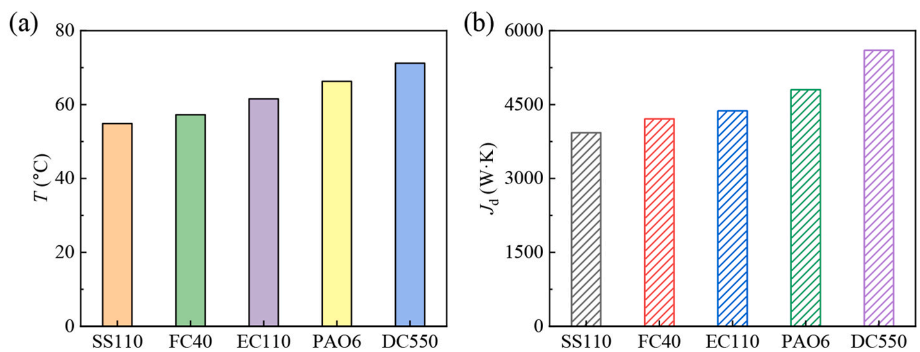

Fig. 12. Comparison of single-phase immersion coolants under identical conditions: (a) center temperature of the heated surface; (b) entransy dissipation.

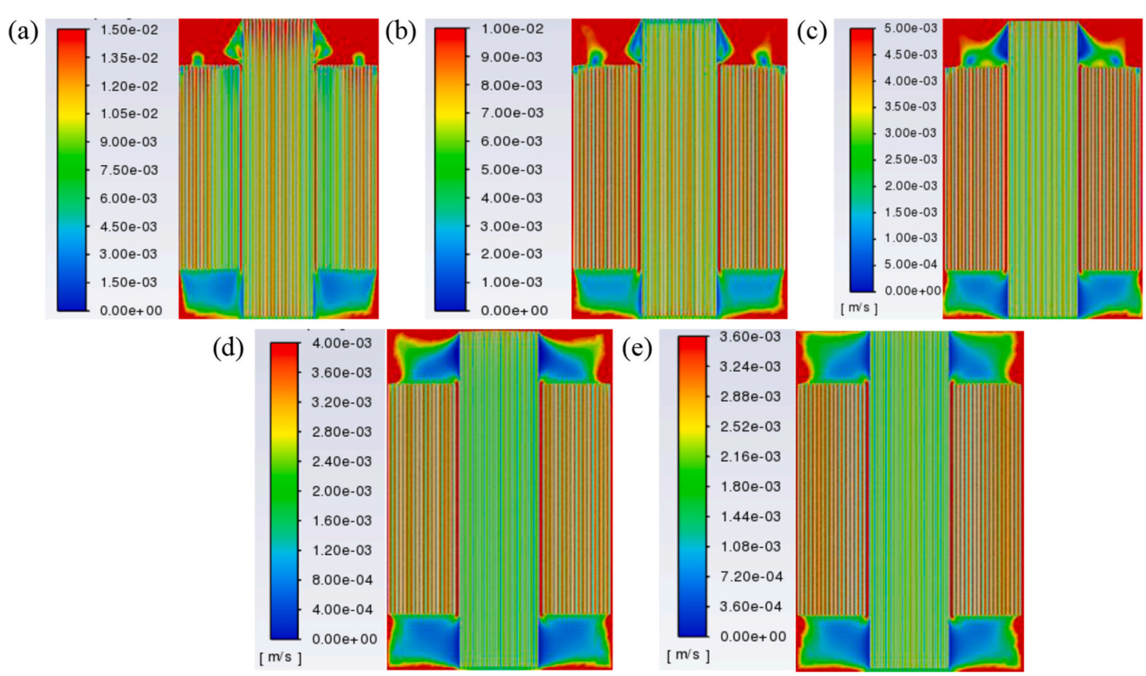

Fig. 13. Velocity contours at a constant height for various coolants (a) SS110; (b) FC40; (C) EC110; (d) PAO6; (e) DC550.

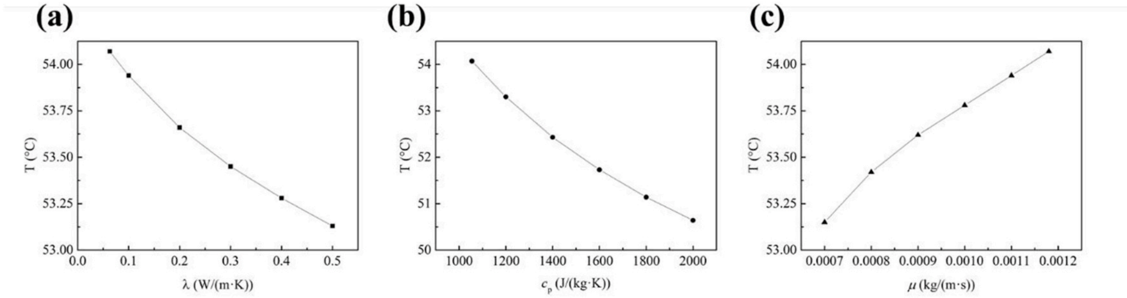

Fig. 14. Evolution of the heated-surface center temperature with the (a) thermal conductivity, (b) specific heat, and (c) dynamic viscosity of the SS110 coolant.

$$J_{\mathrm{d}} = \iiint \kappa (\nabla T)^2 dV \tag{55}$$

Based on the extremum principle of entransy dissipation, minimized dissipation at a constant heat flux reflected lower thermal irreversibility and enhanced heat transfer efficiency. Fig. 12(a) showed that wall temperatures increased in the order of SS110 < FC40 < EC110 < PAO6 < DC550. This trend was consistent with the entransy dissipation hierarchy presented in Fig. 12(b), providing strong empirical support for the theoretical principle.

To investigate the underlying cause, the velocity contours at an

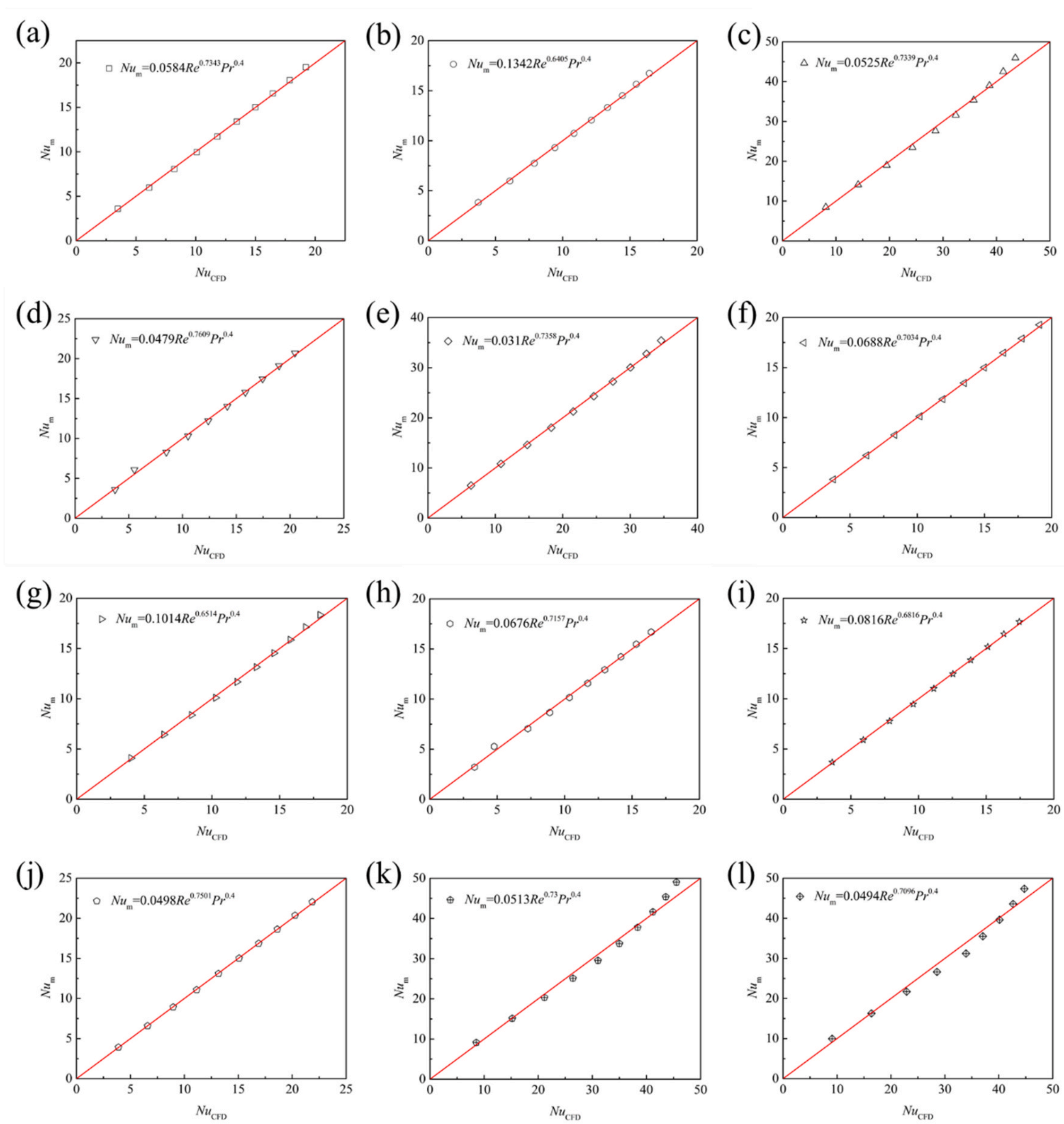

Fig. 15. Comparison of Nusselt numbers from correlations and CFD simulations for various coolants. (a) Euro350; (b) DC550; (c) TMC-40; (d) S5X; (e) SF10; (f) CP 70T; (g) PAO6; (h) PAO4; (i) Mineral Oil; (j) EC-110; (k) FC40; (l) SS-110.

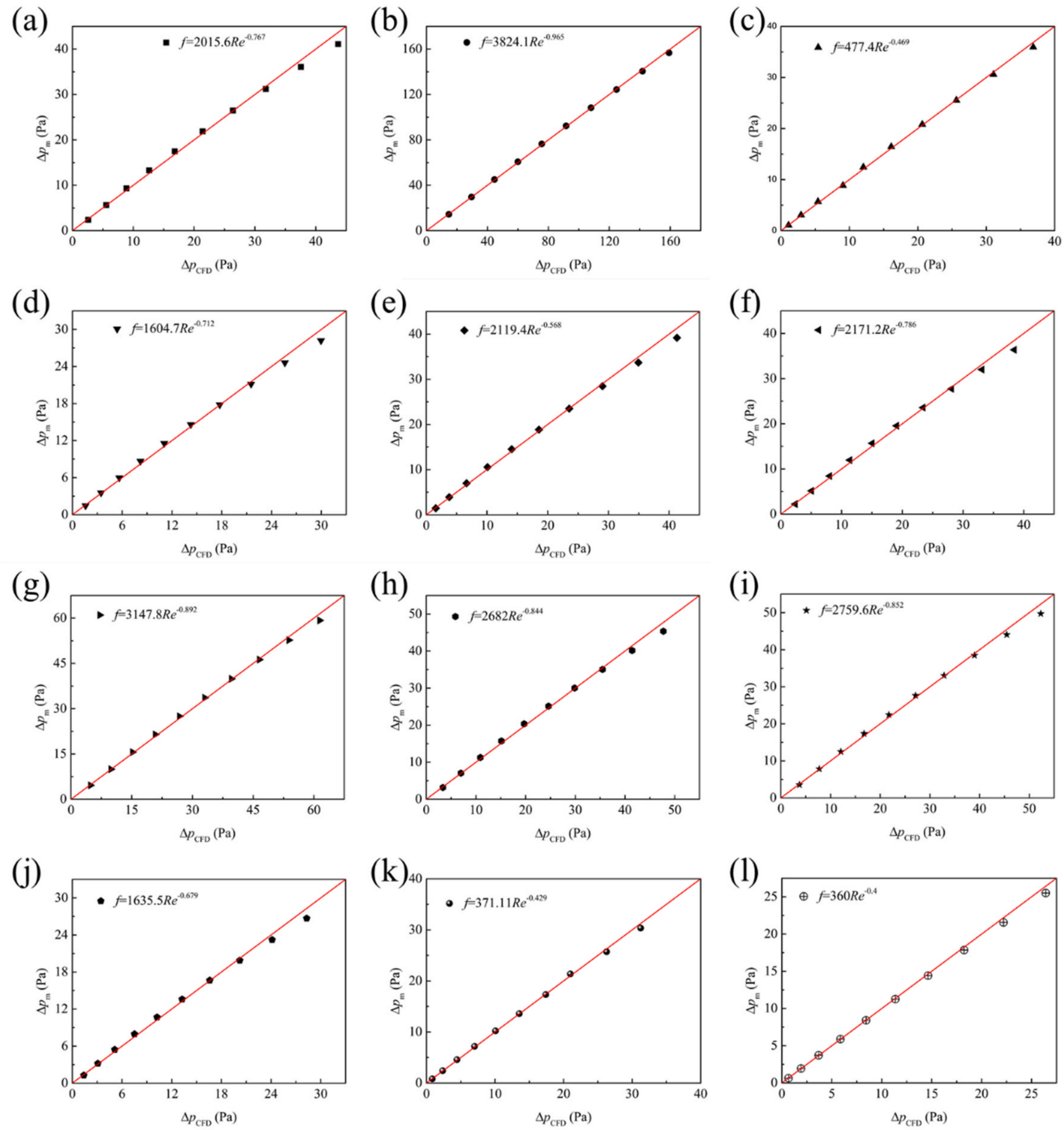

Fig. 16. Comparison of pressure drop from correlations and CFD simulations for various coolants. (a) Euro350; (b) DC550; (c) TMC-40; (d) S5X; (e) SF10; (f) CP 70T; (g) PAO6; (h) PAO4; (i) Mineral Oil; (j) EC-110; (k) FC40; (l) SS-110.

identical height were compared, with a focus on the velocity distribution over the heat sink on the chip. As shown in Fig. 13, the flow velocities for the 5 coolants, in descending order, were: SS110, FC40, EC110, PAO6, and DC550. This variation in flow velocity was the primary reason for the differences in heat transfer performance among the SPIC coolants.

CFD simulations were conducted to investigate the thermal performance of SS110 with varied thermophysical properties. As shown in Fig. 14, increasing thermal conductivity from 0.03 W/(m·K) to 0.5 W/(m·K) reduced center temperature by 0.94 K, while decreasing dynamic viscosity from 0.00118 Pa s to 0.0007 Pa s yielded a 0.92 K temperature reduction. Notably, elevating specific heat capacity from 1038 J/(kg·K) to 2000 J/(kg·K) produced the most significant improvement with a 3.43 K temperature decrease. Hence boosting thermal conductivity and specific heat while reducing dynamic viscosity provided a path to further enhancing the thermohydraulic performance of the SS110 coolant.

3.1.3. Performance metrics for single-phase immersion coolants

Building upon established DC energy metrics PUE and CLF (cooling load factor), we derived the heat dissipation coefficient (HDC) as a performance index integrating thermal-fluid properties of immersion coolants. PUE and CLF can be written as

$$PUE = P_{total} / P_{IT} \quad CLF = P_{cooling} / P_{IT} \tag{56}$$

HDC can be defined as

$$HDC \sim \frac{P_{IT}}{P_{cooling}} = Q \Big/ (q_v \cdot \Delta p) \tag{57}$$

The heat transfer rate Q can be written as

$$Q = hA\Delta T \tag{58}$$

The average Nusselt number (Nu_m) can be calculated as

$$Nu_m = \frac{hL}{\lambda} = f(Re, Pr) \tag{59}$$

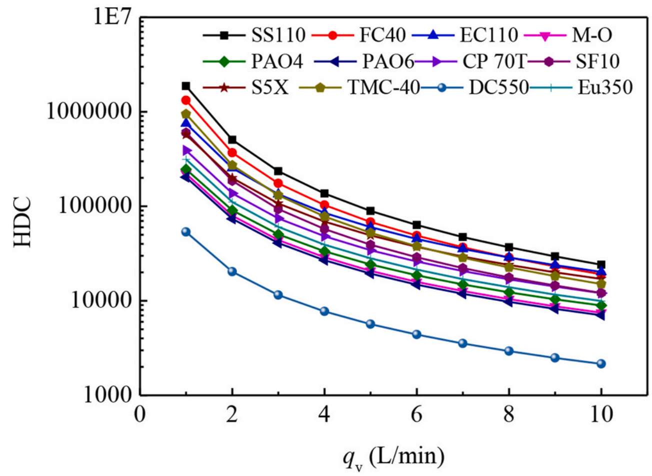

Fig. 17. Comparison of heat dissipation coefficients (HDC) across different coolants.

By comparing the CFD results, we found that the Nusselt number was correlated with Reynolds numbers and the Prandtl number as follows:

$$Nu_m = C_{Nu}Re^{\beta}\,Pr^{0.4} \sim Re^{\beta}\,Pr^{0.4} \tag{60}$$

Here, C_{Nu} was a coefficient dependent on the geometry. For fully developed flow in a pipe, β was 0.8. CFD simulations were performed using the server model proposed by Chhetri et al. [27]. As shown in Fig. 15, equation (60) attained a coefficient of determination R^2 exceeding 0.90, confirming its high accuracy.

The pressure drop can be written as

$$\Delta p = \frac{1}{2} f\rho u^2 L \Big/ d_h \tag{61}$$

The power consumption can be written as

$$P_{\text{cooling}} = \frac{1}{2} f\rho u^3 L \Big/ d_h A_{cs} \tag{62}$$

where A_{cs} was the cross-sectional area.

The friction factor f was governed by the flow regime, making it a complex function of Reynolds number and geometry. In the laminar flow region, $f \sim Re^{-1}$; in the hydraulically smooth region, $f \sim Re^{-0.25}$; and in the hydraulically rough region, $f \sim Re^{0}$. Therefore, a power-law relationship can be assumed between the friction factor f and the Reynolds number, expressed as $f = C_f Re^{-\alpha} \sim Re^{-\alpha}$, where C_f was a coefficient dependent on the geometry. Fig. 16 compared the pressure drops calculated from CFD with those from the correlation, and the good agreement between them validated the proposed hypothesis. The comparison revealed that the C_f for oil-based coolants was significantly higher than that for fluorocarbon-based coolants; specifically, the C_f for DC550 was 10.62 times that of SS110. Equation (57) can be reduced to

$$\frac{Q}{q_v \cdot \Delta p} = \frac{\frac{C_{Nu}\kappa Re^{\beta}\,Pr^{0.4}}{L_c A\Delta T}}{\frac{0.5 f\rho u^3 L A_{cs}}{d_h}} = \frac{C_{Nu}}{C_f}\frac{\kappa^{0.6}c_p^{0.4}\rho^2}{\mu^{2.6}}Re^{\alpha+\beta-3}\frac{2d_h^4 A\Delta T}{A_{cs}LL_c} \tag{63}$$

By decoupling geometric and operational parameters from fluid-specific physical properties, the HDC can be derived as

$$\text{HDC} = \frac{C_{Nu}}{C_f}\frac{\kappa^{0.6}c_p^{0.4}\rho^2}{\mu^{2.6}}Re^{\alpha+\beta-3} \tag{64}$$

Here, C_{Nu} and C_f were coefficients corresponding to differentSPIC coolants and geometries. Fig. 17 compared the HDC for various coolants, and this ranking was consistent with the order of heat removed per unit pumping power. The four fluorocarbon-based coolants (SS110, FC-40, SF10, and TMC-40) significantly outperformed the oil-based alternatives. Specifically, mineral oil, PAO4, PAO6, and DC550 exhibited inferior heat transfer capabilities.

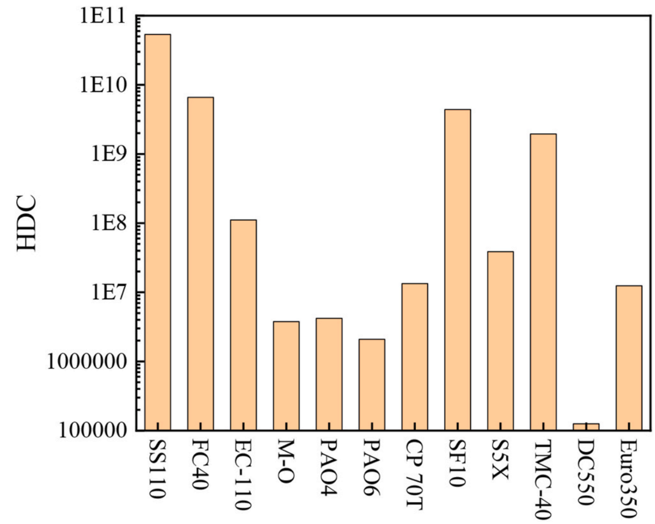

Fig. 18. HDC of various single-phase immersion coolants under equivalent Reynolds number conditions.

Under the condition that $Re^{\alpha+\beta-3}$ was identical, the HDC can be expressed as:

$$\text{HDC} = \frac{C_{Nu}}{C_f}\frac{\kappa^{0.6}c_p^{0.4}\rho^2}{\mu^{2.6}} \tag{65}$$

Fig. 18 presented the HDC of various single-phase immersion coolants calculated from equation (65). The results showed that under equivalent Reynolds number conditions, SS110 provided the optimal thermal performance. Furthermore, equation (65) revealed that the overall heat transfer performance of oil-based coolants diminished as their viscosity increased, consistently lagging behind that of fluorocarbon-based coolants. The proposed framework can be extended to various thermal management systems, including air cooling and cold-plate liquid cooling. Nevertheless, for servers with different architectures, this indicator required integration with CFD simulations or experimental data to ensure the accurate calculation of the HDC value.

Fig. 18 illustrated the HDC of various single-phase immersion coolants, as derived from Eq. (65). The results demonstrated that under equivalent Reynolds number conditions, SS110 exhibited the most superior thermal performance among the candidates. Furthermore, the analytical framework of Eq. (65) revealed an inverse correlation between the viscosity of oil-based coolants and their overall heat transfer efficiency, which consistently underperformed relative to fluorocarbon-based fluids. The proposed framework can be extended to various thermal management systems, including air cooling and cold-plate liquid cooling. Nevertheless, for servers with different architectures, this indicator required integration with CFD simulations or experimental data to ensure the accurate calculation of the HDC value.

3.2. Optimization of heat sink

To mitigate the escalating thermal challenges in high-power servers, a structural optimization strategy was implemented for plate-fin heat sinks, utilizing SS110 as the working fluid due to its superior thermo-hydraulic properties. The baseline configuration comprised plate fins with a thickness of 0.25 mm, spacing of 1.2 mm, and a height of 10 mm. While this spacing was optimized for the low-viscosity SS110 coolant, it would exacerbate thermal resistance if applied to more viscous oil-based coolants. To minimize computational overhead, a single rectangular-fin unit cell was defined as the computational domain (Fig. 2). The

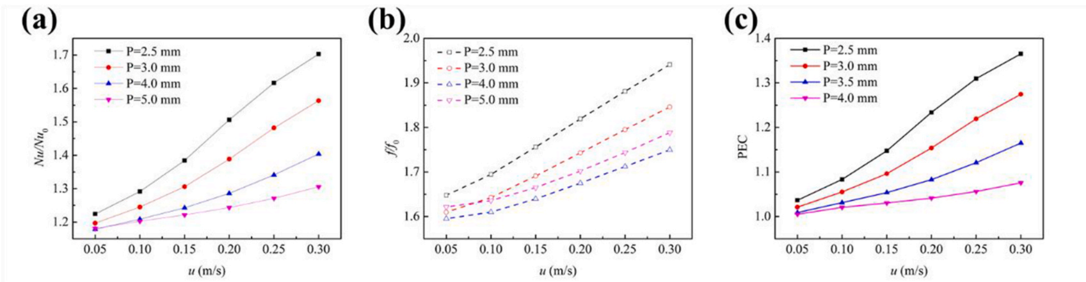

Fig. 19. Variation of (a) Nusselt number ratio (Nu/Nu_0) (b) friction factor ratio (f/f_0) and (c) PEC with the flow velocity and pitch between successive protrusion units at $H_{tu} = 0.2$ mm and H = 10 mm.

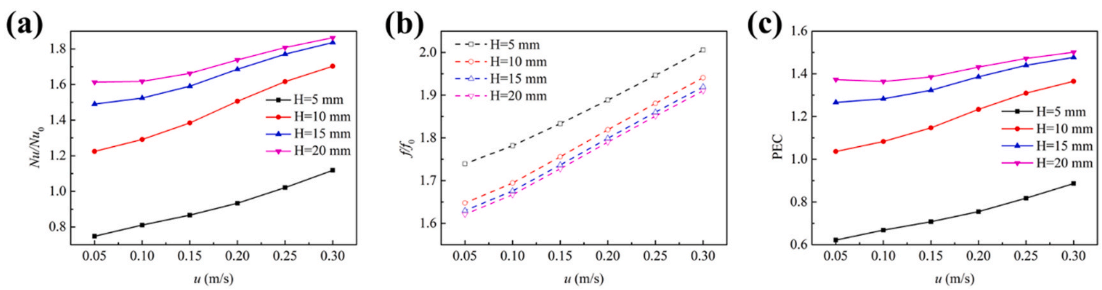

Fig. 20. Variation of (a) Nusselt number ratio (Nu/Nu_0) (b) friction factor ratio (f/f_0) and (c) PEC with the flow velocity and height of fin at $H_{tu} = 0.2$ mm and $L_p =$ 2.5 mm.

optimized design incorporated surface protrusions on the rectangular fins, strategically positioned to induce secondary flow vortices that enhanced convective heat transfer through boundary layer disruption.

Grid independence was rigorously verified using periodic elements with a 5 mm chord length, 5 mm inter-element spacing, and a 0.2 mm protrusion height. The computational domain included 50 mm inlet and outlet extensions to ensure fully developed flow. Numerical simulations were conducted via the SST k-ω turbulence model, selected for its high fidelity in predicting flow separation. The inlet velocity and temperature were 0.3 m/s and 35 °C, while the pressure-outlet condition was applied at the outlet. Non-slip velocity conditions on the walls were applied and the bottom surface was subjected to a uniform heat flux of 20 W/cm^2. As illustrated in Fig. S3, six mesh densities were evaluated. A grid system comprising 1.95 million cells was selected, as the deviations in average bottom-surface temperature and pressure drop were restricted to within 1% and 2%, respectively, compared to the finest mesh.

In the parametric study, the protrusion chord length (L_f) and pitch (L_p) were maintained at a 1:1 ratio to evaluate the influence of L_p on thermohydraulic performance. Fig. 19(a)–(c) depicted the variation of the Nusselt number ratio (Nu/Nu_0), friction factor ratio (f/f_0), and PEC as functions of flow velocity and protrusion pitch. Both Nu/Nu_0 and f/f_0 exhibited a positive correlation with flow velocity, accompanied by a concurrent enhancement in PEC. This suggested that at higher velocities, the convective heat transfer gains significantly outweighed the hydraulic penalties. Reducing the protrusion spacing further augmented both Nu/Nu_0 and f/f_0. Notably, the $L_p = 2.5$ mm configuration achieved a 70% enhancement in convective heat transfer and a 40% improvement in overall thermohydraulic performance relative to the baseline.

The impact of fin height (H) and coolant velocity (u) was summarized in Fig. 20(a)–(c). Nu/Nu_0 scaled positively with fin height, whereas f/f_0 exhibited an inverse relationship across the tested velocity range. Consequently, PEC values improved progressively with fin height, indicating superior performance in taller configurations. Doubling the rib height yielded an 80% increase in Nu/Nu_0 and a 50% increase in PEC compared to the straight-rib geometry, suggesting that protrusion dimensions should be maximized within mechanical constraints.

The influence of protrusion height manifested a non-monotonic relationship with performance indices (Fig. 21(a)–(c)). While small protrusions led to modest increments in Nu/Nu_0 and f/f_0 with velocity, taller protrusions at low velocities resulted in friction penalties that overshadowed heat transfer gains, leading to PEC values below unity. However, at high velocities, the growth in Nu/Nu_0 dominated, restoring the PEC to above 1.0. Optimal performance was observed at $H_{tu} = 0.3$ mm and $u = 0.3$ m/s, where Nu/Nu_0 and PEC reached 225% and 125% of

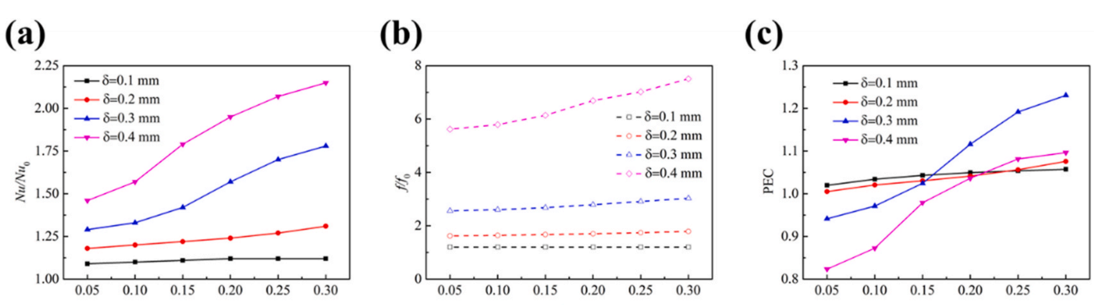

Fig. 21. Variation of (a) Nusselt number ratio (Nu/Nu_0) (b) friction factor ratio (f/f_0) and (c) PEC with the flow velocity and protruding height of the periodic unit at H = 10 mm and $L_p = 5$ mm.

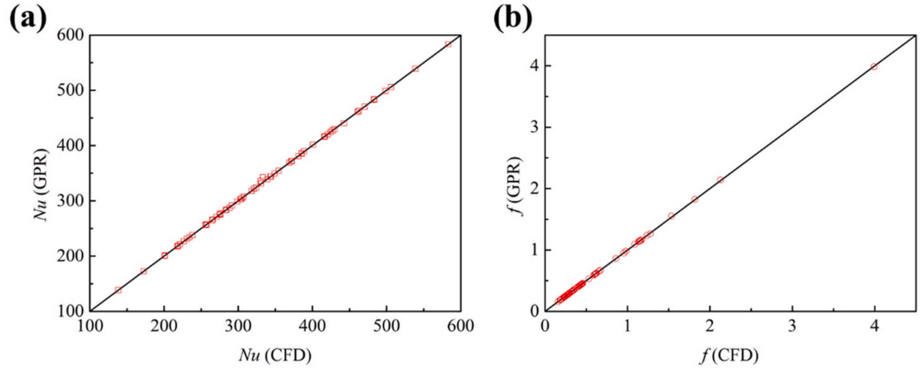

Fig. 22. Comparison of (a) Nusselt numbers (*Nu*) and (b) friction factors (*f*) predicted by Gaussian-process regression (GPR) with corresponding CFD results.

the baseline values, respectively.

The initial intuitive selection of protrusion dimensions was refined through permutation importance analysis (Fig. S5). The results revealed that *Nu* was primarily sensitive to the characteristic size and rib height, whereas *f* was predominantly governed by protrusion size and element density. Leveraging similarity theory, the variables were collapsed into dimensionless groups, and a power-law relationship was established via multivariate least-squares regression.

$$Nu = Re^{0.374}(H/D_h)^{0.437}\left(L_p/D_h\right)^{0.624}(H_{tu}/D_h)^{0.35}(L/D_h)^{0.7}N^{0.65} \tag{66}$$

$$f = Re^{-0.86}(H/D_h)^{-0.057}\left(L_p/D_h\right)^{3.424}(H_{tu}/D_h)^{1.136}(L/D_h)^{-0.02}N^{3.228} \tag{67}$$

The determination coefficients (R2) of developed correlations for Nusselt number and friction factor both exceeded 0.9. Colburn factor *j* and the goodness factor *j/f* were defined as [44].

$$j = \frac{Nu \cdot Pr^{-\frac{1}{3}}}{Re} = Re^{-0.626}\left(\frac{H}{D_h}\right)^{0.437}\left(\frac{L_p}{D_h}\right)^{0.624}\left(\frac{H_{tu}}{D_h}\right)^{0.35}\left(\frac{L}{D_h}\right)^{0.7}N^{0.65}\,Pr^{-\frac{1}{3}} \tag{68}$$

$$j\Big/f = Re^{0.234}\left(\frac{H}{D_h}\right)^{0.494}\left(\frac{L_p}{D_h}\right)^{-2.8}\left(\frac{H_{tu}}{D_h}\right)^{-0.786}(L/D_h)^{0.72}N^{-2.578}\,Pr^{-\frac{1}{3}} \tag{69}$$

Therefore, increasing fin height and protrusion unit density significantly enhanced convective heat transfer coefficient. Structural parameter adjustments simultaneously lead to significant increases in friction factor. Consequently, the fin geometry must be judiciously tailored to the prevailing flow regime.

To improve predictive accuracy for Nusselt number and friction factor, two surrogate models were developed using Machine Learning Toolbox in MATLAB. The CFD-generated dataset was randomly stratified into training (80%) and testing (20%) subsets. Twenty-eight regression algorithms were compared and hyperparameter tuning was performed via Bayesian optimization with 5-fold cross-validation. Gaussian-process regression (GPR) emerged as the most accurate, achieving an R^2 of 0.99 for *Nu* and *f*. The kernel function of GPR for *Nu* and f were rational quadratic function and squared exponential function, which can be expressed as

$$\mathrm{cov}(x_i, x_j|\theta) = \sigma_f^2\left(1 + r^2/2\beta\sigma_l^2\right)^{-\beta} \tag{70}$$

$$\mathrm{cov}(x_i, x_j|\theta) = \sigma_f^2 \exp\left[-\frac{1}{2}\frac{(x_i - x_j)(x_i - x_j)^T}{\sigma_l^2}\right] \tag{71}$$

where σ_f, σ_l, r, and β were the signal standard deviation, characteristic length scale, Euclidean distance between x_i and x_j, positive-valued scale-mixture parameter. While experimental validation of the rib structure was precluded by resource limitations, our findings were substantiated by high-fidelity CFD simulations, a well-established and highly credible method for analyzing single-phase convective heat transfer. The accuracy of our proposed correlation and GPR surrogate model was confirmed by their close alignment with the CFD results, as shown in Fig. 22.

Single-objective optimization of H_{tu} and L_p was performed using a Genetic Algorithm (GA) to maximize PEC across velocities ranging from 0.18 to 0.5 m/s. Particle Swarm Optimization (PSO) and Pattern Search (PS) algorithms provided cross-validation, demonstrating excellent consistency, as shown in Fig. 23(a) and (b). Table S2 listed the corresponding L_p and H_{tu} values under specific flow velocities, which exhibited minimal deviation across GA, PSO, and PS algorithms. Furthermore, the efficiency evaluation criterion (EEC) [45] was analyzed as a function of f/f_0, whose mathematical formulation was expressed as

$$\mathrm{EEC} = \frac{Nu/Nu_0}{f/f_0} \tag{72}$$

Interestingly, EEC attained its peak earlier than PEC, suggesting distinct optimization windows. The single-objective framework achieved a

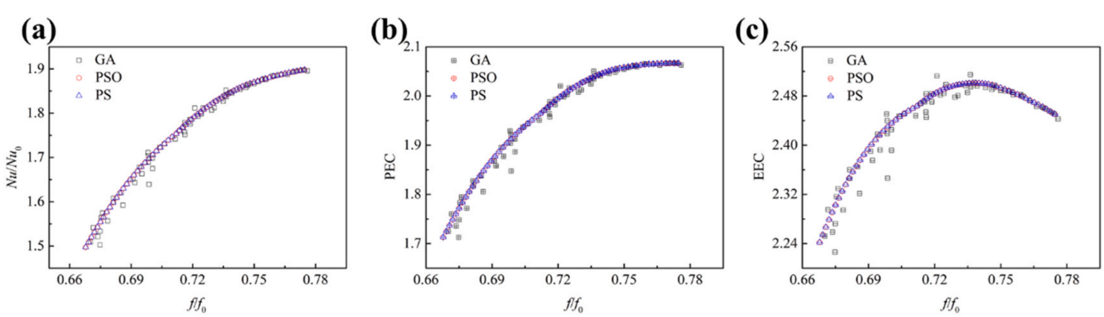

Fig. 23. (a) Ratios of Nusselt number (Nu/Nu_0), (b) PEC, and (c) EEC with the corresponding friction factor ratios (f/f_0) at varying flow velocities yielded by single-objective optimization.

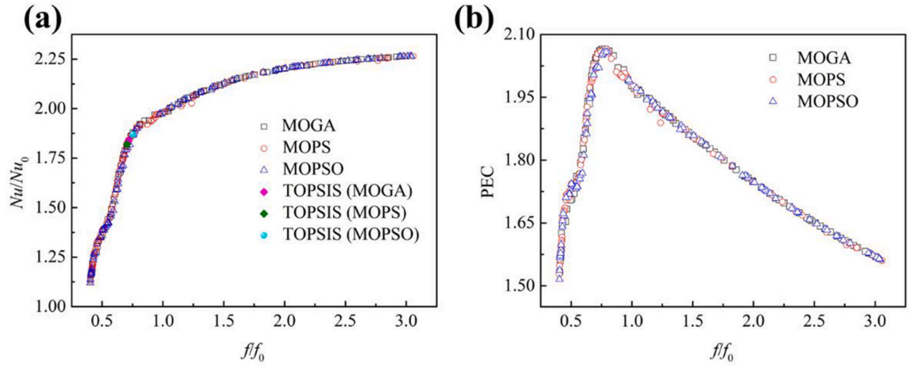

Fig. 24. (a) Ratios of Nusselt number (Nu/Nu_0) and (b) PEC with the corresponding friction factor ratios (f/f_0) yielded by multi-objective optimization.

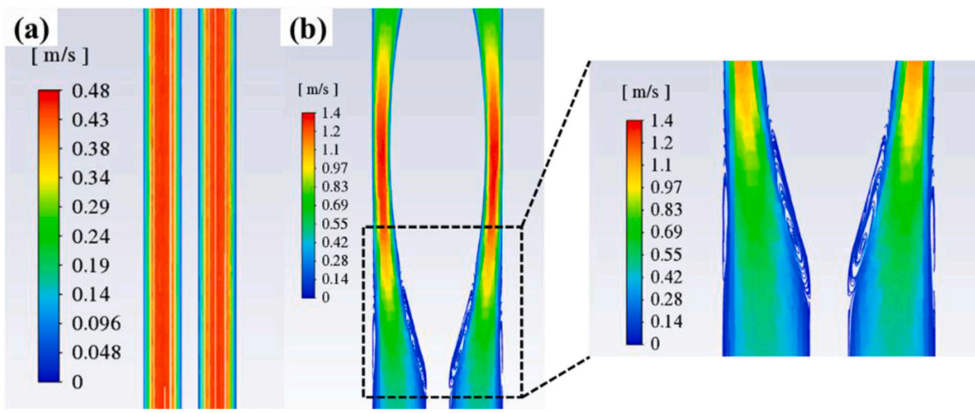

Fig. 25. (a) Streamlines and (b) velocity contours for straight and protruding fins calculated by CFD.

maximum PEC of 2.06 at $f/f_0 = 0.77$.

To resolve the inherent trade-off between heat transfer and pressure drop, a Pareto-based multi-objective optimization (MOGA) was implemented, with Nu/Nu_0 and f/f_0 as competing objectives. This was further validated using MOPSO and MOPS. Table S3 listed key parameter configurations for MOGA and MOPSO. All algorithms generated Pareto fronts composed of noninferior solutions, where each point represented an optimal trade-off between heat transfer and pressure drop. The TOPSIS was subsequently applied to identify compromise solutions balancing heat transfer and pressure drop across the operational range. Fig. 24(a) presented the Pareto front generated by three multi-objective optimization algorithms with the TOPSIS-selected solution. The close alignment of Pareto front across all three methods validated the robustness of MOGA and confirmed convergence in noninferior solution spaces. Fig. 24(b) illustrated the Pareto solutions ranked by PEC, where the TOPSIS method identified a compromise solution achieving PEC = 2.05. The u, H_{tu}, and L_p of Pareto solution determined by MOGA and TOPSIS were 0.444 m/s, 0.2 mm, and 2.6 mm.

Fig. 25 compared the velocity fields of straight fins and protruding fins ($H_{tu} = 0.4$ mm, $L_P = 5.0$ mm) at $u = 0.3$ m/s. The protruding geometry generated vortices near the wall surface, which disrupted thermal boundary layers. Quantitative evaluation of the average synergy angle revealed a consistently larger value for straight fins than for their protruding counterparts.

Finally, as shown in Fig. 26, the proposed heat sink with SS110 coolant significantly outperformed the plate-fin design by Muneeshwaran et al. [21] using FC40 coolant. Our design achieved a 3.83 K reduction in peak chip temperature. The efficacy of the protruding-fin architecture was further validated on a third-generation server (Fig. S6), confirming its ability to provide superior heat transfer enhancement while maintaining manageable pressure drops.

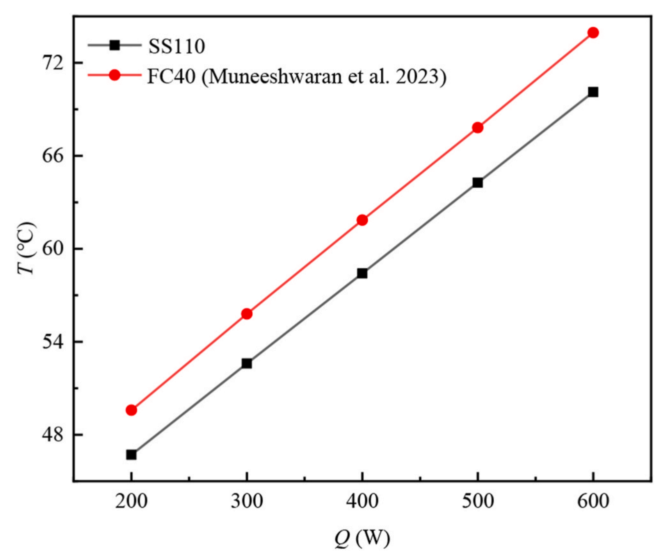

Fig. 26. Comparison of chip surface center temperature between designed heat sink with SS110 coolant and plate-fin heat sink with FC40.

3.3. Experimental comparison between silicone oil and SS110 coolant

A comparative experimental study was performed to evaluate the thermal performance of SS110 dielectric coolant versus silicone oil using the SPIC system (Fig. 4). The experimental setup, detailed in Fig. S7, utilized a customized immersion tank with strategically configured inlet and outlet ports. During the experiments, only the central inlet was active to deliver coolant directly to the heat sink. Temperatures were monitored via T-type thermocouples (±0.5 °C uncertainty), while flow

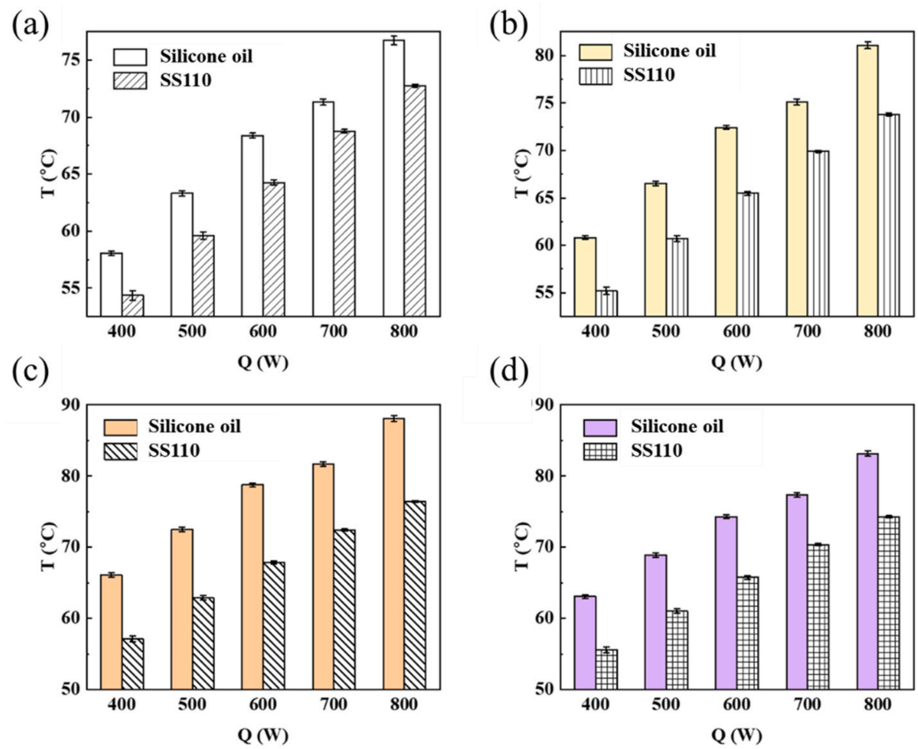

Fig. 27. Measured temperatures at identical locations under the same heating power and volumetric flow rate. (a) Average temperature; (b) Standard deviation of temperature.

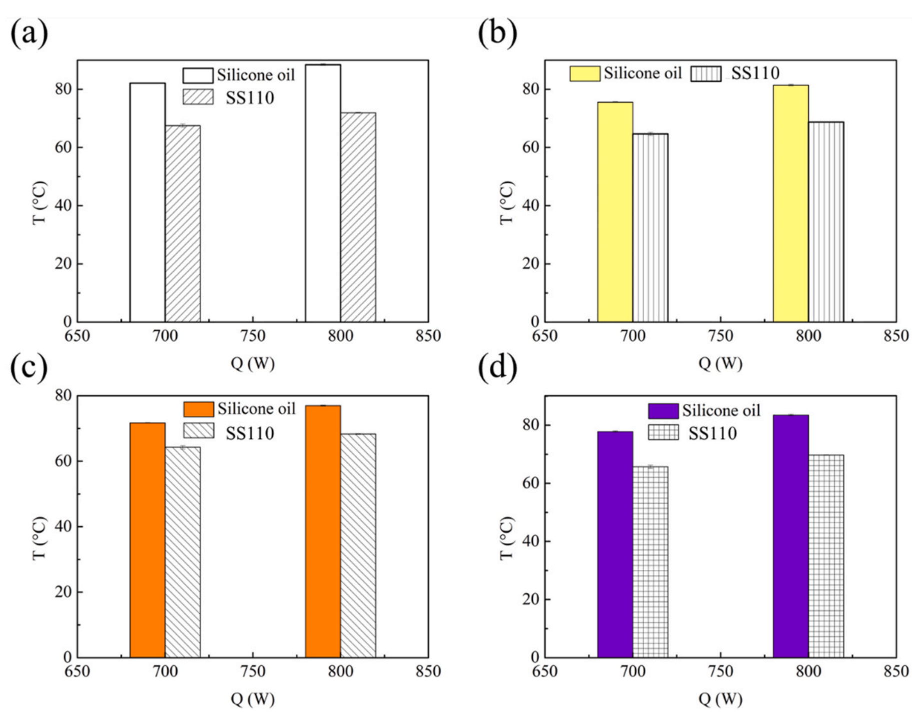

Fig. 28. Measured temperatures at identical locations under the same heating power and pump consumption. (a) Average temperature; (b) Standard deviation of temperature.

rates were regulated by an electromagnetic flowmeter (1% uncertainty). As depicted in Fig. S8, consistent volume flow rates were maintained to ensure comparable operating conditions. Results in Fig. 27 indicated that SS110 coolant achieved a maximum temperature reduction of 11.67 °C compared to silicone oil, accompanied by a 60% improvement in surface temperature uniformity. Notably, the SPIC system maintained a partial Power Usage Effectiveness (pPUE) of approximately 1.03.

Furthermore, the thermal characteristics were evaluated under identical heating power and pumping power consumption. As illustrated in Fig. 28, the application of SS110 coolant reduced the temperature at the same position by up to 16.5 °C and decreased the temperature non-uniformity by approximately 65%. Fig. S9 compared the volumetric

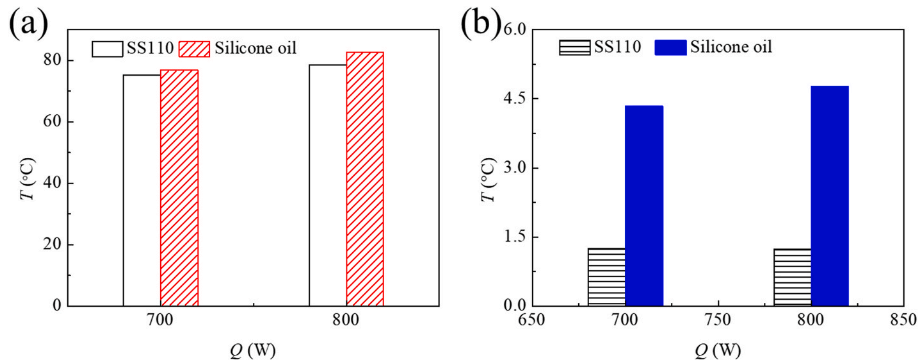

Fig. 29. Measured temperatures at identical locations under the same heating power and mass flow rate. (a) Average temperature; (b) Standard deviation of temperature.

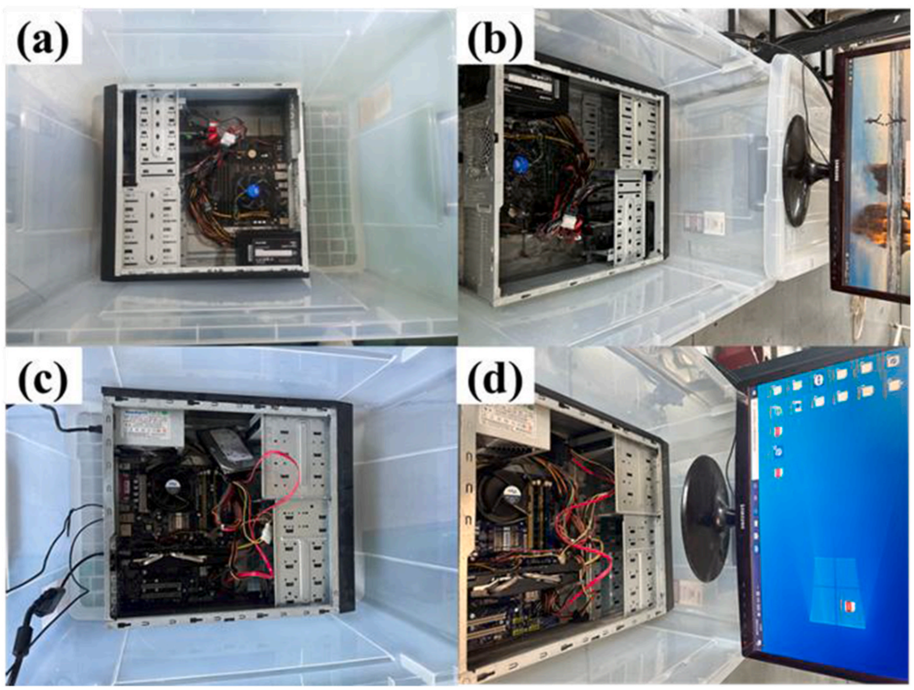

Fig. 30. (a) Server submerged in silicone oil. (b) After submerging in silicone oil for over 1000 h, the server was retested for operation capability. (c) Server submerged in SS110 coolant. (d) After submerging in SS110 coolant for over 1000 h, the server was retested for operation capability.

flow rates, revealing that the SS110 flow rate was approximately 30% higher than that of silicone oil at the same power input. This enhancement was primarily attributed to the significantly lower viscosity of SS110, which reduces flow resistance.

To further isolate the influence of fluid properties, a controlled investigation was conducted at a fixed mass flow rate of 0.2 kg/s (Fig. S10). While the mass flow rate was equivalent, the volumetric flow rate of silicone oil was 87% higher than SS110 due to the smaller density. Despite this hydrodynamic disadvantage, SS110 coolant still demonstrated superior thermal performance. As shown in Fig. 29(a) and (b), SS110 achieved a 1.6 K lower average surface temperature and a 72% reduction in temperature standard deviation. These results underscored the efficacy of SS110 in mitigating localized hot spots and associated thermal stress.

Finally, a rigorous 1000-h immersion compatibility test confirmed the operational viability of both coolants with commercial servers (Fig. 30). However, material compatibility assessments (Fig. S11) revealed that while thermal pads remained stable in both media, silicone grease gradually dissolved in silicone oil. Consequently, thermal pads were identified as the optimal thermal interface material for SS110-based SPIC systems.

3.4. Optimization of the tank structure

In practical applications, multiple servers were typically integrated into a single tank. As shown in Fig. S12, Conventional architectures featuring opposing inlet and outlet ports often suffer from severe flow maldistribution, with flow rate disparities between servers reaching a factor of 20. To address this, a symmetric inlet/outlet configuration was designed based on flow field characteristics, as shown in Fig. 31. With the exception of a 50 mm reduction in the height of the overflow area, the dimensions of the remaining components were identical to those shown in Fig. S12.

The porosity distribution of the perforated plate was found to be a critical determinant of flow distribution. As shown in Fig. 32(a), a uniform porosity distribution resulted in the central server receiving up to 80% less coolant than the peripheral servers, driven by the lower flow resistance near the outlets. To rectify this, a non-uniform porosity strategy was implemented. Specifically, the perforated plates directly beneath the side servers were designed to be impermeable, while the perforated plates beneath the remaining servers were set to a porosity of 0.1. As demonstrated in Fig. 32(b), this optimization constrained the maximum flow disparity to less than 11%, representing an 85%

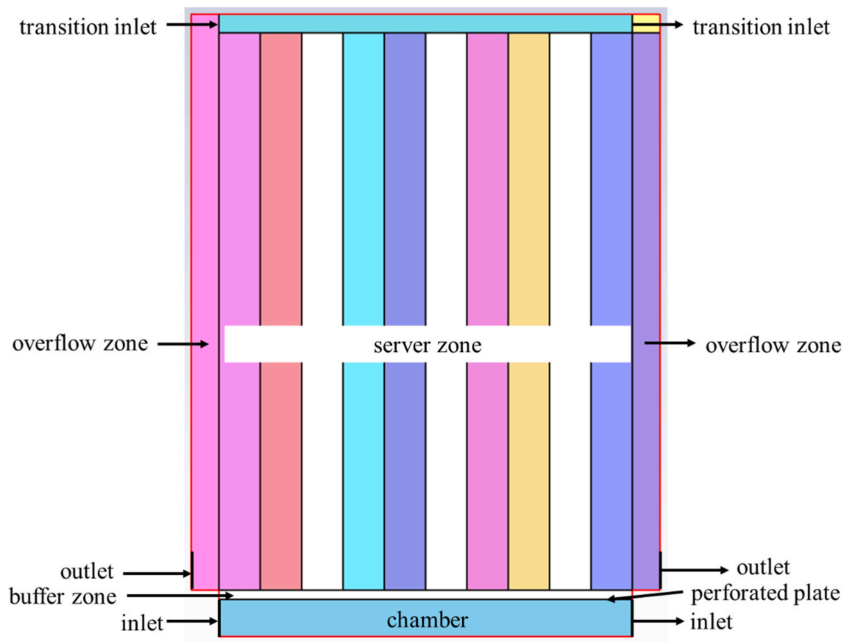

Fig. 31. Single-phase immersion cooling tank with symmetrical inlet and outlet configuration.

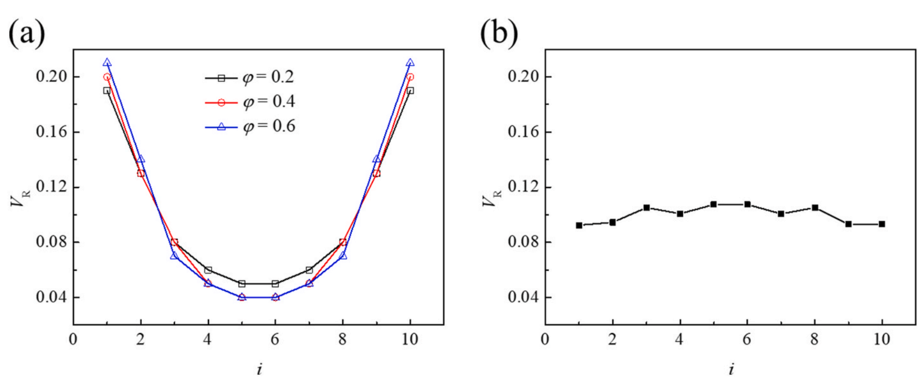

Fig. 32. The influence of the porosity distribution in porous plates on the coolant flow rate received by servers. (a) Porous plate with uniform porosity, and (b) Porous plate with non-uniform porosity.

improvement in coolant distribution uniformity.

4. Conclusion

This study established a comprehensive computational and experimental framework for advancing the thermal management of high-performance servers via SPIC systems. Rigorous analysis highlighted the superior thermophysical properties of the fluorocarbon coolant SS110, which exhibited a viscosity one to two orders of magnitude lower than traditional oil-based coolants. Forced convection simulations indicated that SS110 outperforms FC40 in 1U server configurations, yielding an 11% increase in heat dissipation capacity and a 7% reduction in pressure drop. Furthermore, the introduction of the Heat Dissipation Coefficient (HDC), based on PUE and CLF metrics, provided a robust framework confirming the energy-efficiency potential of SS110, particularly when optimized for low viscosity and high specific heat.

To surpass the limitations of conventional heat sinks, a novel protruding-fin geometry was developed to enhance convective heat transfer through vortex-induced flow modulation. Empirical correlations for the Nusselt number and friction factor were established using similarity theory, demonstrating high predictive fidelity. This was coupled with a GPR surrogate model and multi-objective genetic optimization to derive a Pareto front. The optimal design, selected via the TOPSIS method, achieved a remarkable PEC of 2.05. Furthermore, the tank configuration was optimized to significantly mitigate coolant flow maldistribution, decreasing the non-uniformity of coolant distribution by over 85%.

The theoretical findings were rigorously validated through systematic experimental investigations. Integrating the optimized protruding-fin heat sink with SS110 coolant into a real server environment reduced the chip center temperature by an additional 3.83 K compared to the baseline FC40 with plate-fin heat sinks. Furthermore, under a high heating load of 800 W, SS110 demonstrated a 16.5 °C temperature reduction and a 72% improvement in temperature uniformity relative to silicone oil. Prolonged immersion testing further confirmed the material compatibility and reliability of the system. The implications of this work are significant for the evolution of server thermal management. The proposed HDC metric and the optimized cooling solution provided practical guidelines for reducing PUE and CLF, ultimately contributing to more sustainable and efficient computing infrastructures. Future research could explore the integration of miniature impellers into SPIC system to enhance local convection on high-heat-flux chips. Furthermore, coupling reinforcement learning with variable frequency pumps may offer a pathway to dynamically regulate coolant flow, thereby ensuring the reliable and effective operation of high-power chips.

CRediT authorship contribution statement

Bo Zhang: Writing – original draft, Visualization, Software, Methodology, Investigation, Conceptualization. **Hongrui Li:** Methodology. **Tiexiao Xu:** Investigation. **Lu Wang:** Investigation. **Liang Chen:** Investigation. **Zhen Li:** Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work is supported by the S&T Program of Hebei (Grant No. 20374506D), funding from DAIKIN Industries, Ltd., and the financial support from the Science and Technology Project of State Grid Zhejiang Electric Power Company, Ltd. Jinshuitan Hydropower Plant (Contract No. 20242001533).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.energy.2026.140315.

Nomenclature

Symbol			
C_1, C_2	Constants	p	Pressure, Pa
C_{Nu}	Nusselt Number Correlation Constant	P_{k}	Generation term
c_{p}	Specific heat, J/(kg·K)	q_m	Mass flow rate, kg/s
D_{h}	Hydraulic diameter, m	q_v	Volumetric flow rate, m^3/s
f	Friction factor	q_w	Wall heat flux, W/m^2
f_{app}	Apparent friction factor	r	Euclidean distance between x_{i} and x_{j}
H	Fin height, m	Re	Reynolds number
H_{tu}	Protruding height of periodic unit, m	S_{T}	Heat source term
K_{c}	Contraction loss coefficient	t	Fin thickness, m
K_{e}	Expansion loss coefficient	T	Temperature, K
L	Fin base length, m	T_f	Average fluid temperature, K
L_{ext}	Extension section length, m	T_w	Average wall temperature, K
L^*	Normalized fin length	$\boldsymbol{U}$	Velocity vector, m/s
L_f	Chord length of the arc-shaped fin, m	u_{ch}	Channel velocity, m/s
L_{in}	Inlet section length, m	u_{f}	Free stream velocity, m/s
L_p	Inter-unit spacing, m	W	Fin width, m
Nu	Nusselt number	w_{ch}	Fin space, m
Pr	Prandtl number		

Greek symbol			
α	Constant	λ	Thermal conductivity, W/(m·K)
β	Positive-valued scale-mixture parameter	γ	Synergy angle
σ	Porosity	σ_k	Turbulent Prandtl numbers corresponding to turbulent kinetic energy
σ_ε	Turbulent Prandtl numbers corresponding to turbulent dissipation rate	σ_l	Characteristic length scale
σ_f	Signal standard deviation	k	Turbulent kinetic energy
ε	Turbulent dissipation rate	ρ	Density, kg/m^3
μ	Dynamic viscosity, Pa·s	ν	Kinematic viscosity, m^2/s
∇	Gradient operator	Δ	Laplace operator

Abbreviations

BLT	Boundary layer thickness
CFD	Computational fluid dynamics
CLF	Cooling load factor
CRAC	Computer room air conditioning
DC	Data center
EEC	Efficiency evaluation criterion
GA	Genetic algorithm
GPR	Gaussian Process Regression
GRC	Green Revolution Cooling
HDC	Heat dissipation coefficient
MOGA	Multi-objective genetic algorithm
MOPS	Multi-objective pattern search
MOPSO	Multi-objective particle swarm optimization
PEC	Performance evaluation criterion
PS	Pattern search
PSO	Particle swarm optimization
PUE	Power usage effectiveness
RAM	Random Access Memory

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SPIC	Single-phase immersion cooling
TDP	Thermal design power
TOPSIS	Technique for Order Preference by Similarity to an Ideal Solution
TPIC	Two-phase immersion cooling

Data availability

Data will be made available on request.

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